

Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. "Collateralized Debt Obligation Overview." *Khan Academy*, <https://www.khanacademy.org/economics-finance-domain/core-finance/derivative-securities/cdo-tutorial/v/collateralized-debt-obligation-overview>.
 - **Required pages:** Whole video
 - **Estimated reading time:** 15 minutes
2. "Collateralized Debt Obligation." *Khan Academy*, <https://www.khanacademy.org/economics-finance-domain/core-finance/derivative-securities/cdo-tutorial/v/collateralized-debt-obligation-cdo>.
 - **Required pages:** Whole video
 - **Estimated reading time:** 10 minutes
3. "Mortgage-Backed Security Overview." *Khan Academy*, <https://www.khanacademy.org/economics-finance-domain/core-finance/derivative-securities/mort-backed-secs-tut/v/mortgage-back-security-overview>.
 - **Required pages:** Whole video
 - **Estimated reading time:** 10 minutes
4. "Mortgage-Backed Securities III." *Khan Academy*, <https://www.khanacademy.org/economics-finance-domain/core-finance/derivative-securities/mort-backed-secs-tut/v/mortgage-backed-securities-iii>.
 - **Required pages:** Whole video
 - **Estimated reading time:** 10 minutes
5. Open Risk Manual. "Five Cs Of Credit Analysis." https://www.openriskmanual.org/wiki/Five_Cs_Of_Credit_Analysis
 - **Required pages:** All
 - **Estimated reading time:** 10 minutes
6. Romero-Torres, Jennifer et al. *Securitization in India: Managing Capital Constraints and Creating Liquidity to Fund Infrastructure Assets*. Asian Development Bank, 2017.
<https://www.adb.org/sites/default/files/publication/379076/securitization-india-infrastructure.pdf>
 - **Required pages:** pp. 22–31, 39–48
 - **Estimated reading time:** 60 minutes

NOTE: Content presented in the Required Readings and Lesson Notes will be tested in all assignments, including Quizzes.

SECURITIZATION

Reading Time	90 minutes
Prior Knowledge	Mortgage, Mortgage-backed security (MBS)
Keywords	Investment bank, Special purpose entity (SPE) / Special purpose vehicle (SPV), Collateralized debt obligation (CDO), Tranche, Waterfall cash flow

In Module 4, we defined a derivative as a financial instrument whose value depends on the value of some underlying asset(s). Now, we refine this definition to say that a derivative has a finite lifetime and that its value is known precisely at either inception (e.g., swap) or expiration (e.g., option). Thus, while a weather derivative does not have an underlying asset or security (often called “the underlying”), it still satisfies the price condition at one of the boundaries. We also introduced the concept of a mortgage-backed security (MBS) as a type of derivative security that is backed, or secured, by a bundle of home loans. If you need a refresher on mortgages, refer to Module 4, Lesson 3 as needed.

In this lesson, we track a mortgage from **origination** to **securitization**. Origination is the creation of the mortgage loan itself: a bank originates a mortgage simply by entering a mortgage contract and lending money. Securitization was defined in Module 4 as the creation of new securities that are collateralized by assets such as mortgages. We will also discuss credit risk and how to analyze it conceptually. Then, we discuss in more detail the transformation of mortgages into MBSs via the securitization process.

1. Introduction to Mortgages

Mortgages are used by individuals and businesses to buy real estate without paying the entire purchase price up front. We will refer to either an individual or a business as the borrower. Once approved, the borrower gets the money to buy the real estate unit and repays the loan plus interest over a specified number of years until they own the property free and clear. Typically, mortgage terms are for 15 to 30 years with monthly payments due at the beginning of each month.

Most traditional mortgages are fully amortizing. This means that the regular payment amount will stay the same, but different proportions of principal vs. interest will be paid over the life of the loan with each payment.

If you’ve never seen an amortization table for a fully amortizing mortgage you should take a look at this [spreadsheet](#) and you are invited to play with it.

You should be able to understand why a mortgage contains credit risk for the lender. Essentially, it takes a long time for the borrower to repay the lender. During that time, several things could go wrong. For instance, a borrower could suffer a job loss or a severe illness, which could impair the borrower’s ability to repay the lender; as a result, the lender could lose part of the principal and the interest payments.

While the mortgage rate is fixed in our spreadsheet, mortgages can contain variable rates. Imagine that mortgage interest rates were to go from 2.5% to 7.5%. If you play again with the spreadsheet, you will notice that the monthly total payment can go up substantially, maybe even above what the borrower can afford to pay, in which case the borrower will not be able to make the agreed-upon timely payments. In this case, the lender takes control over the property from the borrower using a legal process called **foreclosure**. Then, the lender will likely try to sell the property, but it could still not raise the money owed to them.

There is also another possibility: the borrower could realize that the mortgage is **underwater**, i.e., the borrower owes more than the home is worth. In some cases, it makes sense to quit making payments and walk away from the mortgage. While this will have severe consequences for the credit score of the borrower, it also puts the lender at risk.

Furthermore, not all mortgages are the same, even aside from the amounts. Some borrowers with low credit ratings are not able to obtain a **prime conventional mortgage** because lenders view them as having a greater-than-average risk of defaulting on the loan. However, these borrowers could qualify for what are called **subprime mortgages**. This class of mortgages often involves adjustable rates (ARM), which allows for the interest rate to potentially increase at a specified point in time during the life of the mortgage. As such, the higher interest rates on subprime mortgages are intended to compensate the lenders for the additional risk.

Specific to the United States, there is also a class of mortgages known as **jumbo loans** or **mortgages**. A jumbo loan is a mortgage used to finance properties that are too expensive for a conventional mortgage. Loan limits are set by the **Federal Housing Finance Agency** (FHFA) and assigned to two government-sponsored enterprises in terms of the maximum amount the FHFA agrees to cover: the Federal National Mortgage Association (FNMA, colloquially known as *Fannie Mae*) and the Federal Home Loan Mortgage Corporation (FHLMC, commonly known as *Freddie Mac*). Note that these are government sponsored enterprises (GSE) and not government agencies. These two GSEs are responsible for buying the bulk of U.S. residential mortgages from lenders, and they set the limit on the maximum value of any individual mortgage that they will purchase from lenders. This amount is important because lenders rely on these purchases to free up liquidity and provide new loans. As of this writing (2024), the limit is \$766,550 in most counties, but it changes every year. Jumbo mortgages may have high credit quality, but they are in amounts above conventional conforming loan limits. In areas of the U.S. where the prices of real estate are very high, most mortgages could be jumbo loans. Traditionally, the interest rates on jumbo mortgages are higher than for conforming mortgages, but not in every case. However, jumbo loans have been associated with some of the craziest excesses during the housing bubble (2004-2007). During that period, over \$3 trillion worth of jumbo loans were originated and offered at terms that were too easy and which ultimately precipitated the collapse that ensued, as most of us know by now.

Once one understands that mortgages are complex creatures and that there are risks behind mortgages, one should also realize why lenders put borrowers through what is usually a rigorous vetting process. We could say that if this process had been equally rigorous during the housing bubble, perhaps we would not have experienced what we did, but that is a story for another time. We will look at how banks and lenders in general vet prospective borrowers more in detail in the next section.

2. Credit Risk

If you are lucky and prudent, you might be able to save enough money for a down payment on a home. In the United States, the down payment amount is typically 20% of the home price. For example, if you want to buy a home for \$500,000, you will need 20% of \$500,000, or \$100,000, as a down payment. You will need to borrow the \$400,000 from a family member or friend or, more likely, from a bank. You could try to take out a personal loan, but more often, at least in the U.S., you would probably take out a mortgage loan. A bank that provides a mortgage is said to **originate a mortgage**.

In the U.S., the borrower applies for a mortgage from a retail bank, or possibly from a mortgage bank that specializes in residential mortgages. (Note that some banks may specialize in commercial mortgages.) Borrowing is a type of credit, and banks are traditionally in the business of providing credit (loans), so it’s no surprise that loans have credit risk. What is credit risk? As you recall from Module 1, credit risk is the risk that the lender does not receive the money that is owed to them in full and on time. In other words, credit risk is the risk of loss due to non-payment. In the case of a mortgage, it is the risk that the borrower does not pay back all of the money borrowed to buy a home.

To manage credit risk, a bank requires a significant amount of credit information from its clients. For instance, the borrower must complete a lengthy application for a mortgage. This application requires various types of information. You can probably already guess at least some of the types of information that a bank would want to know about a borrower before deciding to originate a loan.

Before moving on, take a minute or two to write down what information you would want to know about a borrower before loaning them money.

Then, [check out the Open Risk Manual from the required reading](#). Does what you wrote down match one or more of the 5 Cs of credit analysis?

Next, we’ll take a closer look at the 5 Cs:

- Capacity;
- Capital;
- Character;
- Collateral; and
- Conditions.

2.1 Capacity

Very simply, capacity refers to the ability of the borrower to repay a loan. This means that a lender needs to consider the borrower’s revenues and expenses to decide whether the borrower has the capacity to make the required payments.

To do this, lenders may look at credit scores since a way a person handles credit cards and personal credit can show how the person manages the credit they receive. However, certain metrics play an important role, like debt-to-income ratio (DTI), in which all outstanding debt is compared to a person’s earnings. The lower the DTI, the higher the likelihood that the borrower will be able to make timely payments.

Assume that you are extending a loan to someone called Nushi. You would probably want to know whether Nushi has the capacity to pay you back. To this purpose you may ask “Does Nushi have a job?” If so, she is more likely to be able to pay you back from her paycheck. If she owns a business, is the business profitable? If so, she is more likely to repay the loan with the profits from her business.

You find out that Nushi has a job, so you feel better about loaning her money because having a job increases her capacity to repay her debt. However, you may find out that, despite having a job, Nushi also owes money to the bank, and this could worry you as her income needs to go toward repaying two debts, if she does in fact get the loan from you. These should both be considered in deciding whether to give Nushi a loan.

2.2 Capital

In general, without showing some capital, it is not likely that you will get a loan or a line of credit. If a person has a large net worth (everything they own minus everything they owe), they are more likely to be able to pay you back.

The analysis is the same for a company’s **capital** or net assets, which is equal to total assets minus total liabilities.

Suppose Nushi has a car, a nice television, and a nice watch. She just doesn’t have much cash right now, and she needs to buy gas so she can get to work. She already owes money to the bank, but not that much, so her net worth is still strong: she owns much more than she owes.

2.3 Character

In addition to Nushi’s capacity to pay back the loan and her capital, we must also consider Nushi’s **character**. Consider Nushi’s previous history when it comes to repayment of debt to determine how likely it is that Nushi will be able to repay you.

Is Nushi a responsible person? Has she paid back her loans in the past? If you know that she is the type of person who takes her obligations very seriously, then you can probably expect her to take this new obligation seriously as well. Similarly, if she has a history of borrowing money from you and paying you back, then you are probably more comfortable loaning her money again because her character shows that she is more likely to pay back her next loan.

2.4 Collateral

Collateral refers to the assets a borrower can put up as security. In this situation, we would want to know what collateral Nushi can offer up if she is unable to repay the loan. Note that collateral is not always a requirement when taking out a loan.

Now let’s say that Nushi wants to borrow 3,000 Nigerian Naira (NGN). She says, “Take my watch and hold onto it until I pay you back.” In that case, her watch is the collateral for the loan she wants. You believe her watch is worth 4,000 NGN. You feel more comfortable loaning her 3,000 NGN because if she does not pay you back, you can keep the watch. Alternatively, you can sell the watch for around 4,000 NGN so that you can recoup the money you loaned Nushi.

2.5 Conditions

Conditions, the fifth C, is often the most ambiguous or broad.

Conditions might refer to contractual terms of the loan agreement, such as the principal amount, the repayment period, the interest rate, and the purpose of the loan. You might be willing to loan Nushi 3,000 NGN, but not 300,000 NGN. Additionally, you are willing to loan money to Nushi until she can pay you back in a day or two, but not in a month, a year, or 30 years. You might not loan Nushi 3,000 NGN if she promises to only repay the 3,000 NGN, but perhaps you will if she promises to pay you 5,000 Naira the next day. You might only feel comfortable loaning money to Nushi for gas so she can go to work and then pay you back, meaning you might be willing to loan money to Nushi as long as she keeps her job. But if she quits her job, you may want to be paid back immediately. On the other hand, you might not be comfortable loaning Nushi money for stylish sunglasses or fancy clothes, so you might loan her the money only on the condition that she uses it for an asset rather than a consumable.

Conditions might also refer to the financial or economic environment of the borrower. If economic conditions in the country are currently challenging (like in a recession), then the borrower may have more difficulty repaying the loan. Conditions might also refer to the financial condition of a specific sector or industry. For example, a drought may mean that agricultural conditions are tough, and a farmer may have difficulty paying back a loan. On the other hand, if Nushi works in a restaurant in a city with a robust tourist economy, then economic conditions are more suitable for loaning her money.

3. Mortgage Origination

Now that we’ve discussed basic credit analysis with respect to a relatively small personal loan, you know the types of questions a bank will ask a mortgage applicant before deciding whether to loan money for a home. Banks apply these same concepts to their decisions about providing loans (or any other form of credit) to their corporate clients. Similarly, investors use these concepts to determine which bonds to buy, as bonds are another form of credit very similar to a loan.

Let’s return to the mortgage example from the beginning of this lesson and make some assumptions about how the 5 Cs of credit analysis apply to another hypothetical scenario.

Let’s say Jacob has been running a farm for 10 years and making a profit (capacity). He owns the land that he farms (capital). Jacob is also very conscientious about paying his employees and suppliers whatever he owes them (character). He’s been renting a room in a house, but now he wants a mortgage to buy his own house. The house will be the collateral for this mortgage loan. Jacob can afford to make a down payment of 20% of the house price (\$100,000 over \$500,000) and pay a decent interest rate (good contractual conditions). The weather has been good, and the harvest of Jacob’s crops should be profitable (good financial conditions). In reality, a bank would want many more details, but let’s assume Jacob’s mortgage application was approved. Jacob receives a mortgage loan of \$400,000, which is a \$400,000 debt or liability, and in combination with the down payment of \$100,000, he can now afford to purchase the home. The bank has originated a mortgage, which consists of a corresponding \$400,000 asset.

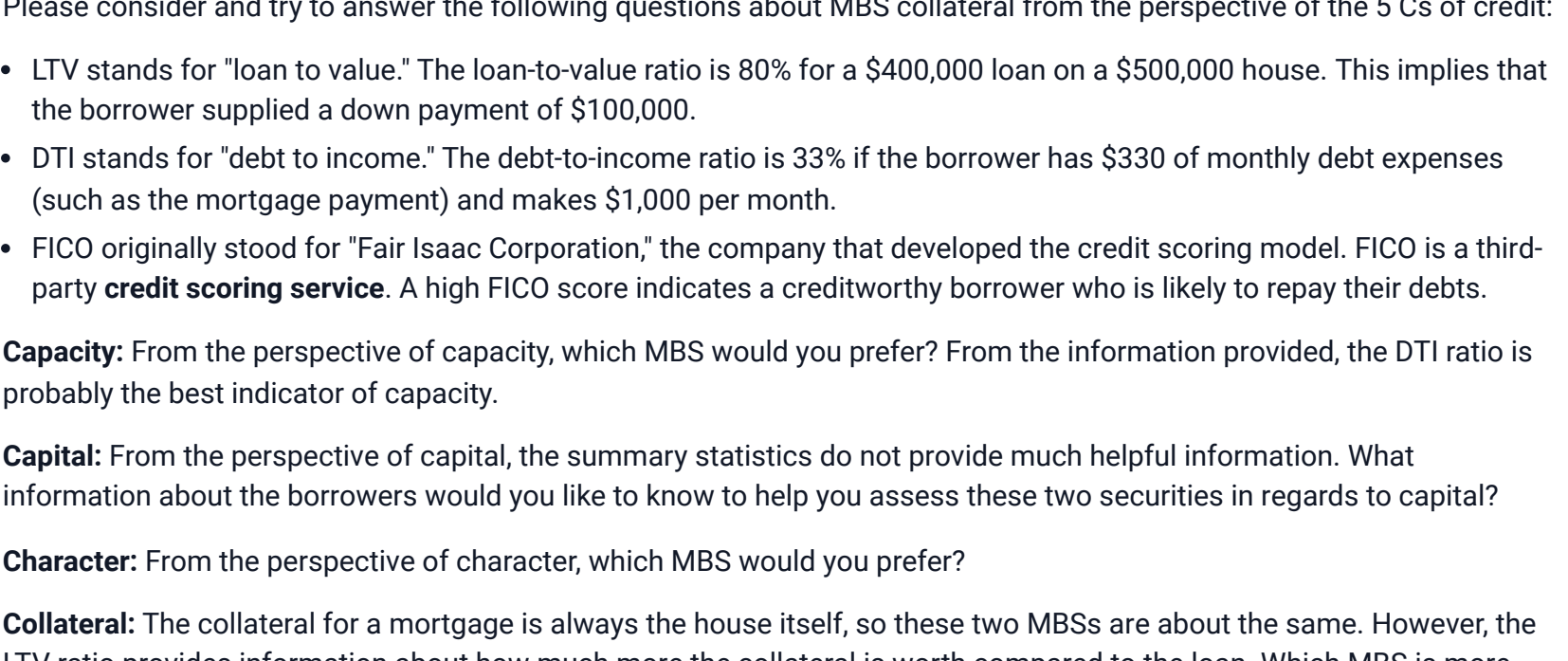
The bank can now hold on to this mortgage loan asset and wait for Jacob to make monthly payments of principal and interest until he has fully paid back the loan. However, there are at least two reasons that this may be undesirable from the bank’s perspective:

1. The mortgage term is likely 30 years, which is a long time for the bank to have its capital tied up in a single transaction. If the bank needs money during that time (say if depositors in the bank want to withdraw money), it could be hard for this bank to sell a single mortgage. A single mortgage is considered illiquid because most investors and other financial players do not want a single mortgage; there isn’t a market for single mortgages. If the bank desperately needed to sell the mortgage loan because it needed money to pay its liabilities (e.g., the depositors want to withdraw their money), another bank might buy the mortgage but for less than it is worth. The risk of not being able to sell an asset at all or having to sell an asset at a steep discount is called **liquidity risk**.
2. This retail bank may not have much capital, and \$400,000 may be a sizable portion of their assets. Let’s consider an extreme example and say that the bank only has \$4 million in capital. That means the bank loaned Jacob 10% of their capital. If Jacob is the only borrower who defaults (doesn’t make the monthly mortgage payments) on the loan, the bank’s capital decreases to \$3.6 million, which may be unacceptable to investors in the bank as well as the bank’s senior management. Too much of their capital, and too much risk, is concentrated in a single borrower. This is known as **concentration risk**. We saw from Module 3 that this is the opposite of a diversified portfolio. Just as we might be interested in having diversification in a stock portfolio and not having too much concentration in a single stock, the bank would also want diversification in the form of mortgage loans to a variety of borrowers.

If the bank does not want to keep the mortgage because of these risks, it can sell Jacob’s loan to an investment bank. What is an investment bank? Why does an investment bank want these mortgages and their concentration and liquidity risks? What will the investment bank do with the mortgages it buys? We’ll answer these questions in the next section.

4. The Investment Bank

Investment banks have a major role in the securitization of mortgages and other assets. But what is an investment bank? An investment bank is different from a commercial or retail bank in that its clients are usually larger institutions, though some of the services provided overlap across bank types.



The following motto summarizes the strategy of investment banks active in mortgage securitization: “We are in the moving business, not the storage business.”

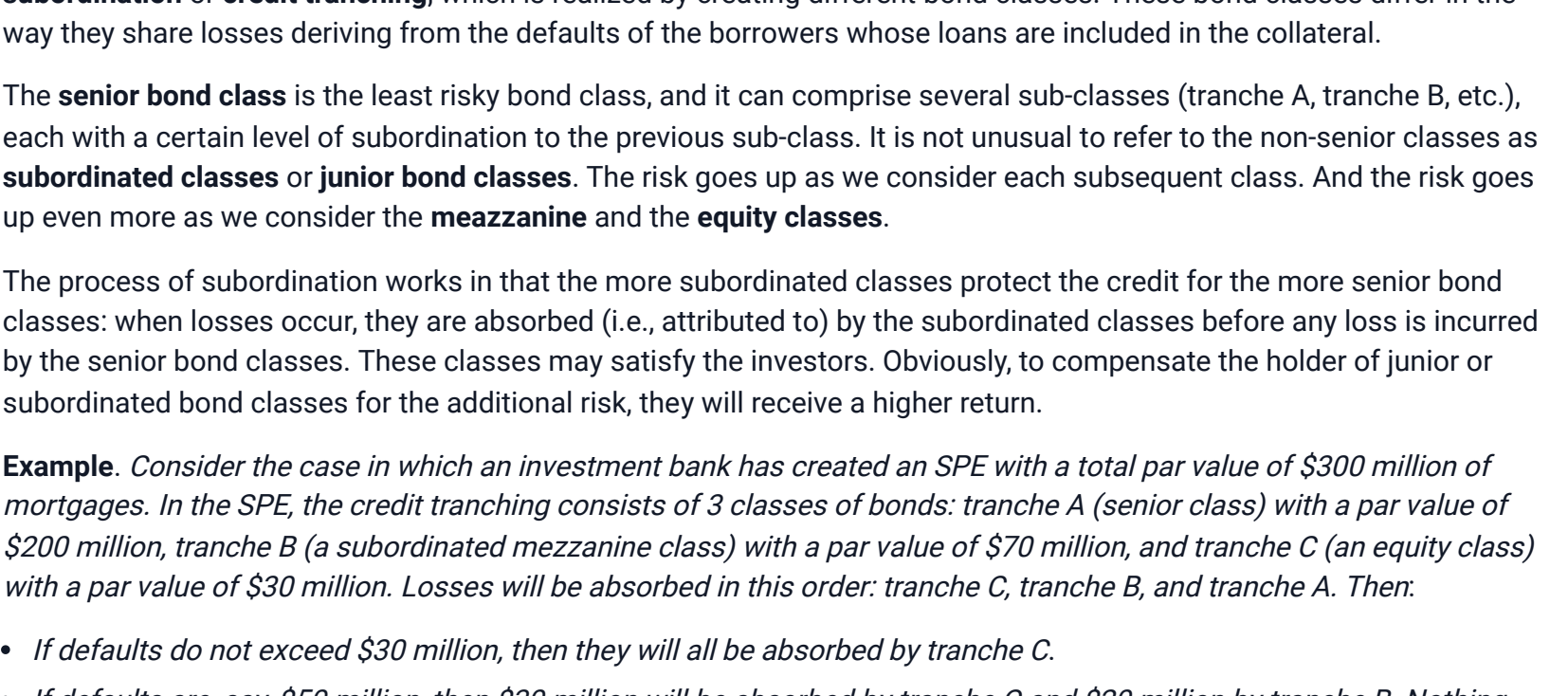
[Watch the “Mortgage-backed Securities Overview” video](#) from the list of required readings and see if you can explain what this motto means in terms of what an investment bank does with mortgages.

Does the investment bank in the video store (hold onto) the mortgages that it buys from the retail bank? Or does the investment bank move (sell) the mortgage into a special purpose entity (SPE)?

Since the investment bank moves these mortgages into an SPE, the investment bank doesn’t keep the liquidity or concentration risks.

[Watch the “Mortgage-backed Securities III” video](#) from the list of required readings and pay close attention to what ultimately happens to all the monthly mortgage payments from borrowers. Where does that money end up?

Figure 1: Process of Mortgage Securitization and MBS Issuance



5. The Securitization Transformation

As you can see, a single mortgage-backed security is backed or collateralized by many thousands of individual mortgages. If an investor buys \$1 million worth of MBSs, a single mortgage will only comprise a fraction of a percent of that investment. The concentration risk we discussed above has been diversified away. The MBS has diversified underlying assets (mortgages) in the sense that there is no concentration in a single mortgage borrower. However, the MBS is still completely concentrated in the larger mortgage market. As such, this concentration in the mortgage asset class cannot be diversified away from MBSs.

Because this concentration risk has been at least partly eliminated, more investors want to buy MBSs; that is, there is an active market for MBSs. As long as there is an active market for buying and selling a security, it is considered liquid. The illiquid, concentrated single mortgage has been transformed into a relatively liquid security with diverse underlying mortgages. We will discuss definitions and calculations related to liquidity in Lesson 4.

We know about the securitization process that results in MBSs. We also discussed the credit concepts used to decide whether to make a personal loan (like to Nushi in our earlier example) or a mortgage loan or when buying a bond. Next, let’s apply these concepts to two MBS investments. Take a look at the summary statistics for the mortgages underlying these two securities.

Please consider and try to answer the following questions about MBS collateral from the perspective of the 5 Cs of credit:

- **LTV** stands for “loan to value.” The loan-to-value ratio is 80% for a \$400,000 loan on a \$500,000 house. This implies that the borrower supplied a down payment of \$100,000.
- **DTI** stands for “debt to income.” The debt-to-income ratio is 33% if the borrower has \$330 of monthly debt expenses (such as the mortgage payment) and makes \$1,000 per month.
- **FICO** originally stood for “Fair Isaac Corporation,” the company that developed the credit scoring model. FICO is a third-party **credit rating service**. A high FICO score indicates a creditworthy borrower who is likely to repay their debts.

Capacity: From the perspective of capacity, which MBS would you prefer? From the information provided, the DTI ratio is probably the best indicator of capacity.

Capital: From the perspective of capital, the summary statistics do not provide much helpful information. What information about the borrowers would you like to know to help you assess these two securities in regards to capital?

Character: From the perspective of character, which MBS would you prefer?

Collateral: The collateral for a mortgage is always the house itself, so these two MBSs are about the same. However, the LTV ratio provides information about how much more the collateral is worth compared to the loan. Which MBS is more attractive from the collateral perspective when you take into account the LTV ratio?

Conditions: One type of condition we discussed was the purpose of the loan. In both of these cases, we know the underlying purpose of the mortgage loans is to buy a house.

We also talked about several contract terms that could impact the decision to provide a loan, such as the term of the loan and the interest rate. How do the underlying pools of loans compare in these regards?

Each summary mentions the average principal balance. Which principal balance would you prefer?

Once you’ve considered your responses to these questions, you can find the answers at the end of the lesson ¹.

6. Tranches and the Payment Waterfall

One of the simplest examples of securitization would involve the sale of only one class of bonds. But this would not necessarily be the best way to repackaging the mortgages or any asset-backed security in general. Investors have different desires for returns and different appetites for risk.

For these reasons, the structure of securitization is often more complicated and involves several classes of bonds. [In the introductory video, “Collateralized Debt Obligation Overview”](#) and [in the “Collateralized Debt Obligation \(CDO\)” video](#) that follows it (both from the required readings), pay special attention to the differences between investment, mezzanine, and equity tranches of the collateralized debt obligation (CDO), especially in terms of risk and priority of cash flows.

From those readings, it should emerge that securitization involves forms of credit enhancement via what the industry calls **subordination** or **credit tranching**, which is realized by creating different bond classes. These bond classes differ in the way they share losses deriving from the defaults of the borrowers whose loans are included in the collateral.

The **senior bond class** is the least risky bond class, and it can comprise several sub-classes (tranche A, tranche B, etc.), each with a certain level of subordination to the previous sub-class. It is not unusual to refer to the non-senior classes as **subordinated classes** or **junior bond classes**. The risk goes up as we consider each subsequent class. And the risk goes up even more as we consider the **mezzanine** and the **equity classes**.

The process of subordination works in that the more subordinated classes protect the credit for the more senior bond classes: when losses occur, they are absorbed (i.e., attributed to) by the subordinated classes before any loss is incurred by the senior bond classes. These classes may satisfy the investors. Obviously, to compensate the holder of junior or subordinated bond classes for the additional risk, they will receive a higher return.

Example. Consider the case in which an investment bank has created an SPE with a total par value of \$300 million of mortgages. In the SPE, the credit tranching consists of 3 classes of bonds: tranche A (senior class) with a par value of \$200 million, tranche B (a subordinated mezzanine class) with a par value of \$70 million, and tranche C (an equity class) with a par value of \$30 million. Losses will be absorbed in this order: tranche C, tranche B, and tranche A. Then:

- If defaults do not exceed \$30 million, then they will all be absorbed by tranche C.
- If defaults are, say, \$50 million, then \$30 million will be absorbed by tranche C and \$20 million by tranche B. Nothing happens to tranche A, which would be fully repaid.
- However, if the SPE were to experience a massive default, say \$140 million, then tranche C and tranche B would be completely wiped out, and tranche A will also suffer a \$40 million loss as the subordinated tranches could only shield it from a loss of up to \$100 million.

While defaults are certainly feared as they could wipe out investments, there is another type of risk that is important to consider: **prepayment risk**. What is prepayment risk? In general, prepayment risk refers to the risk that the principal amount (or at least a portion of the principal amount) outstanding on a loan is prematurely paid back. Why should we worry about prepayment, since we get our money back earlier instead of later? Well, we should think of why someone who owes us money is willing to pay us back earlier than agreed.

Suppose you had a mortgage on which you pay an interest rate of 6%. Now, overall interest levels are going down, and you are offered the opportunity to **refinance** your mortgage with a different bank for just 4.5%. What would you do? Unless there are restrictive clauses in your mortgage agreement or large penalties to pay, you would refinance your mortgage, right? The bank that originally provided your mortgage loan receives its money back ahead of time and can use that money to originate new mortgages. However, and this is where the problem lies, the new mortgage will be for around 4.5% instead of 6%. This is bad news for the bank, and it is usually not good news for the MBS investors either since investors receive their principal back ahead of time and need to reinvest it at what are likely lower rates of return. So, yes, there is less credit risk, but now they face reinvestment risk. To alleviate the issue of the uncertainty of cash flows that occurs with prepayment risk, investment banks can create an SPE whose MBSs consist of bond classes that possess different expected maturities. This type of tranching is called **time tranching**. This works like credit tranching essentially, but the difference is that prepayments are first absorbed by the subordinated classes and then, only after the buffering of the subordinated classes is completely used up, the more senior classes. It is quite unusual for MBSs to trail investors from prepayments as it would be too costly for the investment bank that issues the MBSs. However, time tranching addresses these issues. Of course, again, junior tranches would be compensated with a higher return for assuming more reinvestment risk.

You may wonder: why do we need a special legal entity, the SPE, in securitization? There are several reasons. The SPE will get a rating, as you learned from the required readings/videos. And this rating may be different from that of the company that issues the bonds associated with the securitization process. Using an SPE, the bonds can be backed by specific collateral instead of a generic collateral, which is very relevant in the unfortunate case of bankruptcy. We won’t elaborate on this topic, but the creation of SPEs to issue MBSs is very relevant for legal reasons. While the process can vary based on different jurisdictions, in the case of securitizations, courts have no discretion to change seniority for the very simple reason that the bankruptcy of an issuer does not extend to the SPE.

Now that you have seen how assets can be securitized to create new securities, read the relevant pages of [the Romero-Torres primer on securitization structure in India](#)². The concepts are the same, but this reading goes into a little more detail. You’ll read more about cash flows and other relationships between the participants, from the borrowers to the investors, as well as some additional market players that have special roles to play, such as the servicer, the rating agency, and the credit enhancement provider.

7. Conclusion

In this lesson, we discussed how securitization transforms mortgages into MBSs. When determining the likelihood that a borrower will repay a loan, banks and other lenders consider the 5 Cs of credit: capacity, capital, character, collateral, and conditions. We also discussed how these fundamental credit concepts that determine whether someone gets a loan are applied to credit instruments of all kinds, including MBSs. Investment banks use the 5 Cs of credit when considering concentration of risk. We saw how credit risk can be reapportioned into different levels appropriate for different types of investors and that credit risk can be mitigated by different types of credit enhancement, which can be internal and/or external. We also learned about the different roles in the securitization process, from the mortgage borrower and lender, through the arranger and issuer, to the investor, the rating agency, and the servicer. In future lessons in this module, we will build on these concepts by focusing on the structuring of securitized products such as MBSs and how this results in tailored risk profiles and cash flow timing.

Footnotes

1. Answers:

Capacity:	You would prefer a lower DTI as an indicator of higher capacity to repay.
Capital:	You would like to know how much cash the borrower has on hand (in a safe at home or in the bank), what other types of investments the borrower has, and how much each of these is worth.
Character:	You would prefer the MBS with the higher average credit score, which suggests that the borrower is more likely to repay the loan.
Collateral:	You would prefer a lower average LTV, which implies lesser loan values compared to higher collateral values.
Conditions:	You would prefer shorter-term loans to longer-term loans so that you get paid back sooner. You would also prefer higher interest rates to lower interest rates so that you get paid back more.

1. In this reading about securitization in India, the local bank in the example above plays the role of the originator and lender.

References

- “Mortgage-Backed Security Overview (Video).” *Khan Academy*, <https://www.khanacademy.org/economics-finance-domain/core-finance/derivative-securities/mort-backed-secs-tut/v/mortgage-back-security-overview>.
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Required Readings

Required Readings for this program are open access, which means you should be able to access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. Ashcraft, Adam B., and Til Schuermann. "Understanding the Securitization of Subprime Mortgage Credit." *Federal Reserve Bank of New York Staff Reports, 2008*, Staff Report no. 318.
https://www.newyorkfed.org/research/staff_reports/sr318.html.
 - **Required pages:** pp. 14–16
 - **Estimated reading time:** 20 minutes
2. Eisinger, Jesse, and Jake Bernstein. "From Dodd-Frank to Dud: How Financial Reform May Be Going Wrong." *Business Ethics*, 3 June 2011. <https://www.propublica.org/article/from-dodd-frank-to-dud>.
 - **Required pages:** All
 - **Estimated reading time:** 30 minutes
3. Konczal, Mike et al. *Doomed to Repeat: Debunking the Conservative Story about the Financial Crisis and Dodd-Frank*. Roosevelt Institute, 6 June 2017. <https://rooseveltinstitute.org/wp-content/uploads/2020/07/RI-Doomed-to-Repeat-Dodd-Frank-201706.pdf>.
 - **Required pages:** pp. 32–53
 - **Estimated reading time:** 45 minutes
4. Mišanková, Mária et al. "Determination of Default Probability by Loss Given Default." *Procedia Economics and Finance*, vol. 26, 2015, pp. 411–417, <https://www.sciencedirect.com/science/article/pii/S2212567115008151>.
 - **Required pages:** All
 - **Estimated reading time:** 30 minutes

NOTE: Content presented in the Required Readings and Lesson Notes will be tested in all assignments, including Quizzes.

VALUATION CHALLENGES: MARKET FRICTIONS AND MODEL RISK

Reading Time	75 minutes
Prior Knowledge	Subprime mortgage, Option (put and call) pay-off
Keywords	Collateral pool, First and second lien, Moral hazard, Refinance, Cash-out refinance, Market friction, Model risk, Loss Given Default (LGD), Probability of Default (PD), Recovery Rate (RR)

In the previous lesson, we learned the fundamental 5 Cs of credit analysis and applied these to various examples of credit, including MBSS.

In this lesson, we take this kind of analysis further by applying the 5 Cs to a mortgage pool underlying an MBS. We also learn how the structure of the market can make it hard to get reliable credit information, even from a rating agency. We discuss credit concepts such as probability of default and loss given default, which can be used to model the value of MBSs. Given the complexity of MBS modeling, we need to rely more on rating agencies and other market participants, and we ponder government responses to some of the frictions in the market: we look closely at regulations that came after the subprime mortgage crisis and their effectiveness.

1. The 5 Cs Return

The required reading "[Understanding the Securitization of Subprime Mortgage Credit](#)" lists five credit characteristics of the subprime mortgage, numbered below for easy reference (Ashcraft & Schuermann 14-16):

- Two or more 30-day delinquencies in the last 12 months, or one or more 60-day delinquencies in the last 24 months;
- Judgment, foreclosure, repossession, or charge-off in the prior 24 months;
- Bankruptcy in the last 5 years;
- Relatively high default probability as evidenced by, for example, a credit bureau risk score (FICO) of 660⁺ or below (depending on the product/collateral), or other bureau or proprietary scores with an equivalent default probability likelihood, and/or,
- Debt service-to-income ratio of 50 percent or greater; or, otherwise limited ability to cover family living expenses after deducting total debt-service requirements from monthly income.

Recall the 5 Cs of credit analysis and consider how you might use them to evaluate these characteristics.

The first four all suggest a deficiency in the borrower's *character* or *capacity*: Multiple delinquencies (late payments) in the past year or two most likely means the borrower was unwilling or unable to pay on time. Same with bankruptcy and "judgment, foreclosure, repossession, or charge-off", which are actions taken when a borrower never repaid their debt, as opposed to just repaying late. As discussed in the last lesson, a bad (low) credit score is also interpreted as a *character* issue. Lastly, we recognize the "debt service-to-income ratio" as the DTI ratio we discussed in the last lesson, which is another measure of *capacity*.

In "the motivating example" of the same section, the mortgage pool underlying a particular MBS offering is summarized by, among other data, the following (Ashcraft & Schuermann 14-15):

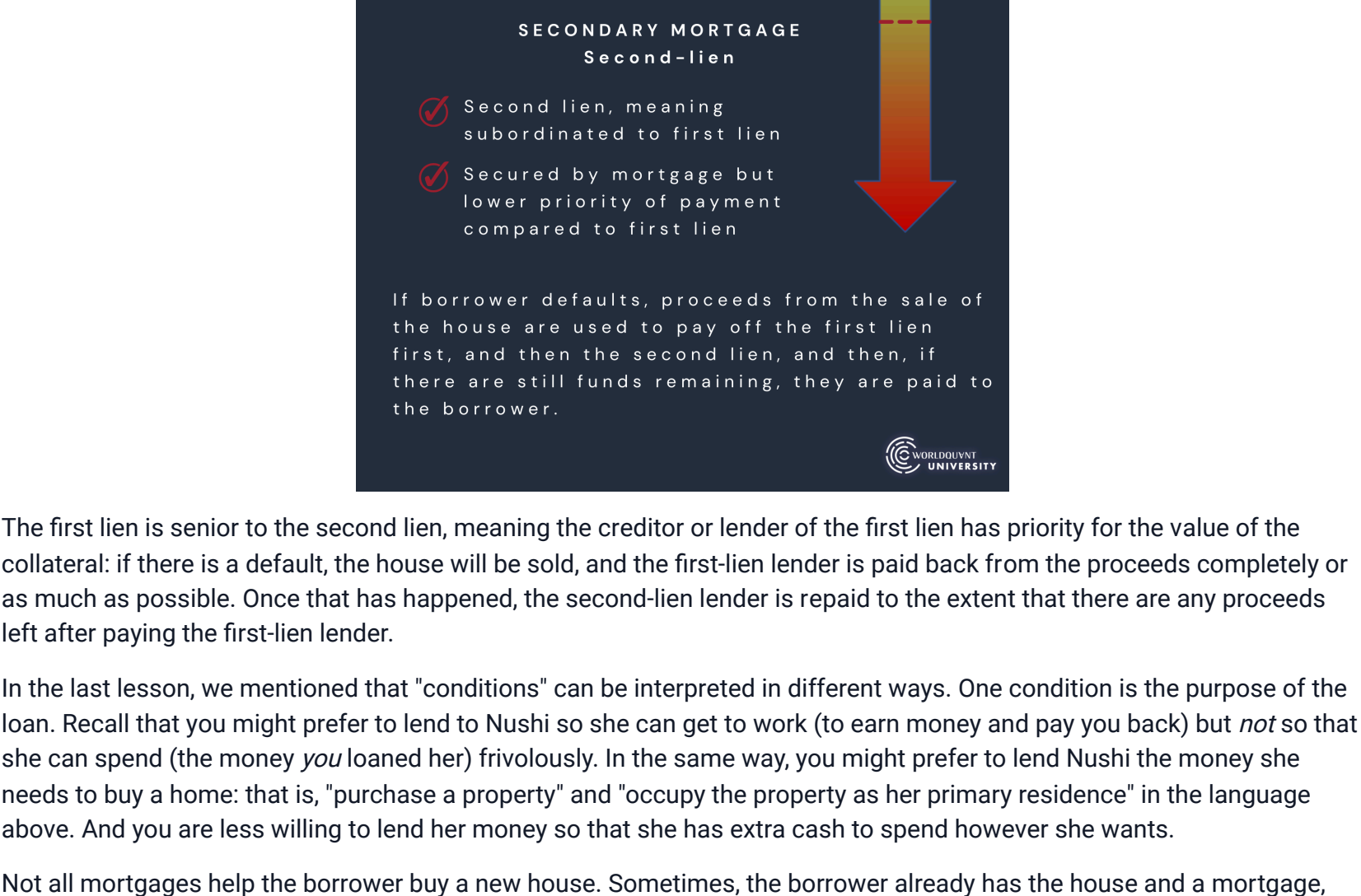
- 98.7% of the mortgage loans are first-lien. The rest are second-lien home equity loans.
- 43.3% are purchase loans, meaning that the mortgagor's stated purpose for the loan was to purchase a property. The remaining loans' stated purpose are cash-out refinances of existing mortgage loans.
- 90.7% of the mortgagors claim to occupy the property as their primary residence. The remaining mortgagors claim to be investors or purchasing second homes.
- The **combined loan-to-value ratio (CLTV)** is the sum of the original principal balance of all loans secured by the property to its appraised value. The average mortgage loan in the pool has a CLTV of 80.34%, which implies an average downpayment of 20%. However, given that this is an average number, the astute credit risk analyst should pay attention to the proportion of borrowers with more than 80% CLTV.

Let's take a look at some other data relating to LTV and DTI in the charts below. Observe how the mortgage default rates ("non-performance probabilities") are highly correlated with LTV ratio and credit score.

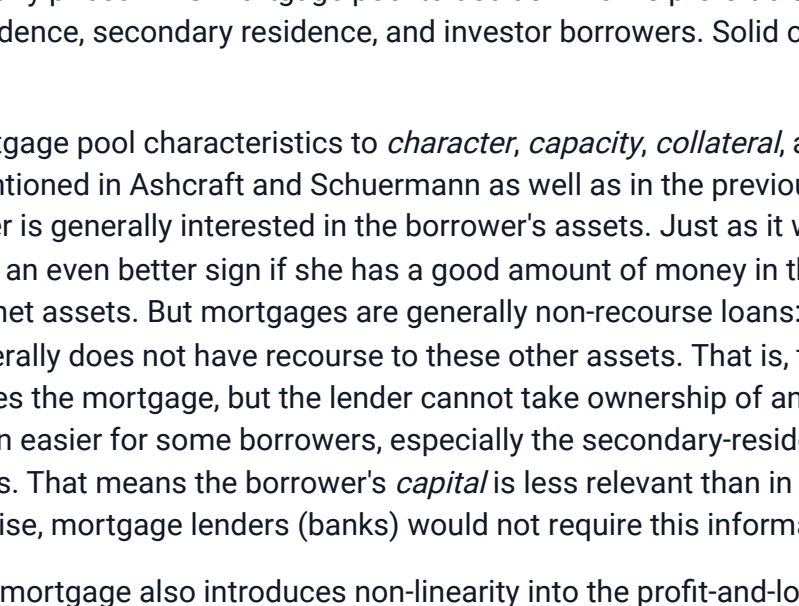
Two notes about these graphs: 1. The description refers to these as "*predicted* non-performance probabilities," but they are not so much predictions or forecasts as they are smooth lines fit to the actual (empirical) data. 2. "NHG" stands for "Nationale Hypotheek Garantie," which is insurance provided by a Dutch government-owned fund, in many ways similar to Fannie Mae or Freddie Mac in the United States or the Japan Housing Finance Agency. The distinction between NHG and non-NHG is not relevant for our current discussion; what is relevant is the upward slope in both graphs representing significant positive correlation between LTV and default and between DTI ("Debt service to income" below) and default.

As you can see, a high LTV, especially as it approaches 100 (equivalent to the loan amount equal to the house value, meaning no down payment), is bad because it suggests a higher probability of default. As you move from left to right (i.e., the LTV increases), the PD increases. Very similarly, the probability of default increases as DTI increases.

Figure 1: Default Rates by LTV and DTI



A first-lien loan has priority over a second-lien loan. (From the perspective of the five Cs, the implications of this difference might be captured under "Collateral"; alternatively, they might be classified under "Conditions" because whether the loan has a first- or second-lien is a "contractual term of the loan.") If a borrower has a second-lien loan and the borrower defaults, then the lender of the first-lien loan is repaid from the proceeds of the liquidated collateral first. The lender of the second-lien loan only benefits if the value of the collateral (a house in this case) exceeds the amount of the first-lien loan. As such, for the MBS investor reviewing the mortgage pool referenced above, the fact that almost all of the loans are first-lien is good: this *collateral* is better than a similar pool with a higher proportion of second-lien loans.



The first lien is senior to the second lien, meaning the creditor or lender of the first lien has priority for the value of the collateral: if there is a default, the house will be sold, and the first-lien lender is paid back from the proceeds completely or as much as possible. Once that has happened, the second-lien lender is repaid to the extent that there are any proceeds left after paying the first-lien lender.

In the last lesson, we mentioned that "conditions" can be interpreted in different ways. One condition is the purpose of the loan. Recall that you might prefer to lend to Nushi so she can get to work (to earn money and pay you back) but *not* so that she can spend (the money you loaned her) frivolously. In the same way, you might prefer to lend Nushi the money she needs to buy a home: that is, "purchase a property" and "occupy the property as her primary residence" in the language above. And you are less willing to lend her money so that she has extra cash to spend however she wants.

Not all mortgages help the borrower buy a new house. Sometimes, the borrower already has the house and a mortgage, and they take out a new mortgage to replace the old one. This is called a **refinancing**: it allows the borrower to reset their mortgage terms, often so that they get a better (lower) interest rate. There are different ways to refinance depending on the reason for the refinance. For example, a **cash-out refinance** is for the borrower who has already repaid some of the principal on an existing mortgage. By replacing the existing mortgage, the borrower can effectively withdraw that principal that they had paid and do whatever they want with the cash-out amount. As with Nushi, this may be an undesirable *condition* for an underlying mortgage.

Similarly, you may not want to lend money so that a borrower can buy a second home. Since it's not where they live as their primary residence, the borrower may be more careless about repaying the mortgage. After all, if they lose their second home, at least they still have their first home. By the same token, you may not want to lend money to an investor who is buying a property so that they can resell (flip) it at a higher price. Since the investor already has a home, they may not care as much about the mortgaged property; if something goes wrong with their investment (a natural disaster or a downturn in the economy), the investor could just walk away by making no further mortgage payments and defaulting on the mortgage.

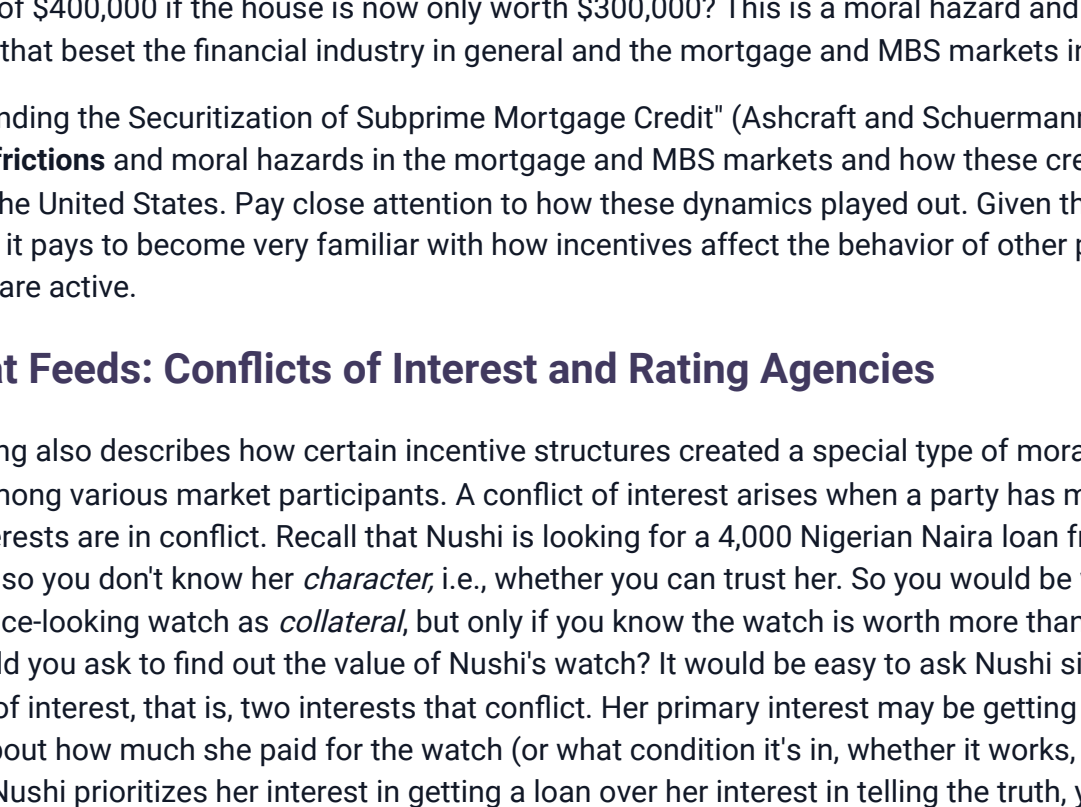
As this relates to the mortgage pool characteristics above, you can compare the percentage of purchase loans (43.3%) to the percentage in another similarly priced MBS mortgage pool to decide which is preferable. And you can do the same with the percentages of primary residence, secondary residence, and investor borrowers. Solid credit analysis of MBSs takes these *conditions* into account.

So far, we have related the mortgage pool characteristics to *character*, *capacity*, *collateral*, and *conditions*. That leaves one C unmentioned: *capital*. As mentioned in Ashcraft and Schuermann as well as in the previous lesson's discussion of the mortgage application, the lender is generally interested in the borrower's assets. Just as it was a good sign that Nushi had a nice TV and a nice watch, it is an even better sign if she has a good amount of money in the bank and/or a decently sized securities portfolio, i.e., strong net assets. But mortgages are generally non-recourse loans: even though these (net) assets are a good sign, the lender generally does not have recourse to these other assets. That is, the lender can take ownership of the house, which collateralizes the mortgage, but the lender cannot take ownership of any *other* property belonging to the borrower. This makes it even easier for some borrowers, especially the secondary-residence and investor borrowers, to walk away from their mortgages. That means the borrower's *capital* is less relevant than in the context of a recourse loan, but it is still meaningful; otherwise, mortgage lenders (banks) would not require this information.

The non-recourse nature of the mortgage also introduces non-linearity into the profit-and-loss profile of the loan, similar to the non-linearity we saw in the discussion of profit and loss from option trades in Module 4, Lesson 1. The relevant graphs from that section are presented again for you below to conveniently visualize the similarity in the risk and reward between options and mortgages.

Say, for example, a bank provides an \$80,000 loan to a borrower to buy a \$100,000 house (the borrower made a \$20,000 down payment): This is analogous to the bank having a short put with an \$80,000 strike. A few months later, the local real estate market deteriorates, and the house is now worth \$80,000. In this situation, most borrowers will continue to make their mortgage payments and avoid default. Though the house price decreased significantly, the bank has not lost any money—yet.

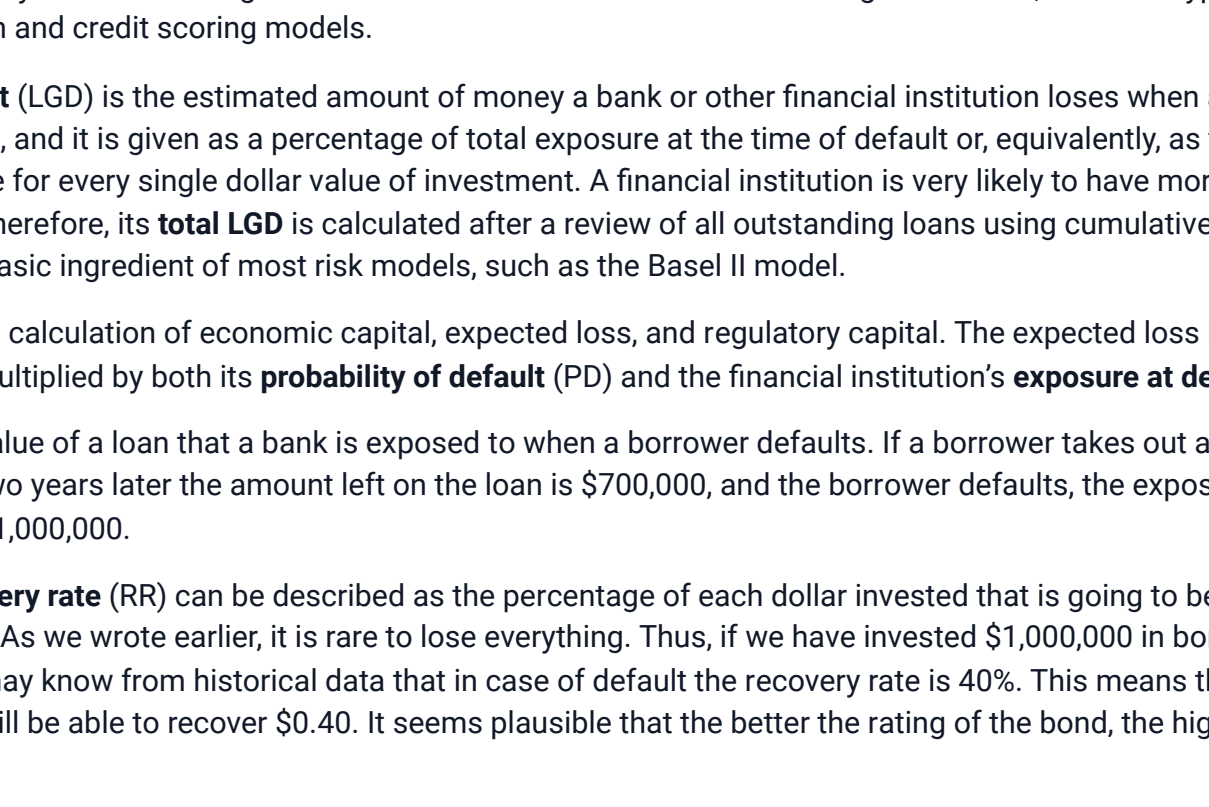
Figure 3: Short Put Payoff Diagram



Now, say the real estate market sinks even more, and the house is only worth \$50,000. (Such a loan is called "under water" because the home value is worth less than the outstanding loan amount.) At this point, many mortgage borrowers will decide to walk away, as described above. If the borrower does default, the bank suddenly has a loss of at least \$30,000 (check the graph): the bank's P/L has moved down the diagonal into a loss. This broken line demonstrates the non-linearity of risk to the bank.

By this same logic, the borrower has a long call position on her house. If her house price increases substantially, she can sell her house, pay back the mortgage loan, and make a profit. Her P/L profile is shown in the graph below: she has all of the upside (potential profit from house price appreciation) and hardly any of the downside (risk of loss).

Figure 4: Long Call Payoff Diagram



2. NINJAs, Liars, and Information Asymmetry

As discussed in the readings, more aggressive, or risky, underwriting allows for more relaxed documentation standards. These loans are also called "no doc" or "low doc" loans, referring to the level of documentation. And here, documentation simply refers to the credit applicant providing proof (documents) of such things as the borrower's income and assets. The borrower may answer the lender's questions about income and assets but *doesn't* have to confirm or prove their answers. For example, the borrower for a **SIVA** mortgage has merely **stated** their **income** but **verified** their **assets**. A **SISA** mortgage borrower has merely **stated** their **income** and **stated** their **assets**. These types of loans had been called "liar loans" because of the ease and frequency with which borrowers could exaggerate their *capacity* and/or *capital* in a mortgage application. Lenders of such loans would rely heavily on the borrower's credit score as an indicator of *character* and overall creditworthiness.

For SIVA, SISA, and **NINA** (**no income** and **no assets** information is provided by the borrower) loans, the lender might still at least check that the borrower is in fact employed where they say they are employed. They are called the **NINJA** loan: **No income** and **no job** or **assets** information is provided by the borrower to the lender. The NINJA borrower is silent on these *capacity* and *capital* characteristics.

3. Information and Moral Hazard

"Informational friction" or "asymmetric information" describes a situation where one party to a transaction or agreement has more or better information than the other. The information is not symmetric across the two parties. For example, the borrower under a "no doc" or "low doc" loan knows how much they have in assets and how much they make in income, but the lender does not have any certainty about these figures.

Asymmetric information easily leads to moral hazard: the party with better information may be tempted to act in bad faith because the consequences are not consequential enough. Recall that the low-doc or no-doc mortgages were called "liar loans." Moral hazard also describes a situation where one party takes excessive risk for their own benefit because they will not bear the full cost of those risks.

The classic example of moral hazard comes from the insurance industry: once a party or person is insured, they may take more risks than before they were insured. If someone has auto insurance, they may not be as careful to put out the fire in their fireplace at night. If someone has insurance against fire theft, they may be more likely to leave their keys in the car or leave their car parked in a bad neighborhood. And similarly, if there is little or no downside to overstating one's income and assets so that person can get a mortgage loan, they will be tempted to do so. After all, if they default on the mortgage because they can't afford to make the payments, the mortgage is non-recourse, so they can just walk away. A person might also choose to walk away if real estate prices decline significantly: why should someone keep making mortgage payments on a principal amount of \$400,000 if the house is now only worth \$300,000? This is a moral hazard and only one of the several moral hazards that beset the financial industry in general and the mortgage and MBS markets in particular.

The reading "Understanding the Securitization of Subprime Mortgage Credit" (Ashcraft and Schuermann) discusses several **informational frictions** and moral hazards in the mortgage and MBS markets and how these created the conditions for the credit crisis in the United States. Pay close attention to how these dynamics played out. Given the profit motive of most financial players, it pays to become very familiar with how incentives affect the behavior of other participants in whichever market you are active.

4. The Hand That Feeds: Conflicts of Interest and Rating Agencies

For example, the reading also describes how certain incentive structures created a special type of moral hazard called "conflict of interest" among various market participants. A conflict of interest arises when a party has more than one interest, and these interests are in conflict. Recall that Nushi is looking for a 4,000 Nigerian Naira loan from you. Say you don't know Nushi well, so you don't know her *character*, i.e., whether you can trust her. So you would be willing to lend her the cash against her nice-looking watch as *collateral*, but only if you know the watch is worth more than the amount you are lending. Who should you ask to find out the value of Nushi's watch? It would be easy to ask Nushi since it's her watch, but she has a conflict of interest, that is, two interests that conflict. Her primary interest may be getting the loan from you. Telling you the truth about how much she paid for the watch (or what condition it's in, whether it works, etc.) may be only a secondary interest. If Nushi prioritizes her interest in getting a loan over her interest in telling the truth, you as the lender will only have yourself to blame because you did not take into account Nushi's conflict of interest. Better to avoid the moral hazard of tempting Nushi to exaggerate (lie) about the collateral value and find another independent source for this information. Independent in this context means finding someone without a conflict of interest: Nushi's boyfriend Obie and Nushi's mother Amara are not independent; they want Nushi to get what she wants and perhaps they even benefit themselves if Nushi can get this loan.

For the same reason, you would not rely on information from Obie or Amara about Nushi's *character* because of their conflicts of interest. And yet, many creditors and other investors rely on information from rating agencies, despite the fact that rating agencies are literally paid by the issuer of the debt they are rating. (The reading "The 2007–2009 Financial Crisis: An Erosion of Ethics" provides details on how all of these properties, and more, made that crisis inevitable.) As you read more about the ease and reliability of rating agencies in Box 3 of the Banco de Mexico's "Financial System Report 2007," consider to what extent you should feel comfortable leveraging their analyses.

5. Model Risk: "All models are wrong but some are useful."

This aphorism attributed to the British Statistician George Box (1919-2013) may oversimplify things, but it makes the point that no one model is correct in every way. Nevertheless, if we make judicious use of models, considering carefully what they purport to tell us and being careful about the inputs and assumptions, then they are indeed useful.

When there is a possibility of default, it is rare to lose everything, and the amount one can lose also depends on whether the loan/investment is collateralized or not. To understand the necessary concepts, we start by introducing some terminology.

Probability of Default (PD) is a term used to measure the likelihood that a borrower will default on their debt obligations within a specific time frame. It is a critical component in credit risk assessment and is crucial in credit risk modeling and more. PD is typically estimated using statistical models and historical data. Among the models, the most typical are logistic regression and credit scoring models.

Loss given default (LGD) is the estimated amount of money a bank or other financial institution loses when a borrower defaults on a loan, and it is given as a percentage of total exposure at the time of default or, equivalently, as the amount that we could lose for every single dollar value of investment. A financial institution is very likely to have more than one investment, and therefore, its **total LGD** is calculated after a review of all outstanding loans using cumulative losses and exposure. It is a basic ingredient of most risk models, such as the Basel I model.

LGD is used in the calculation of economic capital, expected loss, and regulatory capital. The expected loss is calculated as a loan's LGD multiplied by both its **probability of default (PD)** and the financial institution's **exposure at default (EAD)**.

EAD is the total value of a loan that a bank is exposed to when a borrower defaults. If a borrower takes out a loan for \$1,000,000 and two years later the amount left on the loan is \$700,000, and the borrower defaults, the exposure at default is \$700,000 not \$1,000,000.

The **default recovery rate (RR)** can be described as the percentage of each dollar invested that is going to be recovered in case of a default. As we wrote earlier, it is rare to lose everything. Thus, if we have invested \$1,000,000 in bonds with a given rating, we may know from historical data that in case of default the recovery rate is 40%. This means that for every \$1 invested, we will be able to recover \$0.40. It seems plausible that the better the rating of the bond, the higher the RR will be.

This leads us to finally begin to describe LGD via at least two formulae:

$$LGD \text{ (dollar amount)} = EAD \times (1 - RR).$$

and

$$LGD \text{ (as percentage)} = 1 - \frac{\text{Potential Sale Proceeds}}{\text{Outstanding Debt}}.$$

The second formula is useful when there are forms of collateral. Note that the RR has to be estimated, so the LGD estimated with the first formula could seem more technical; however, it might actually sell a formula as more mathematical than it is in reality.

Example. A bank has loaned \$400,000 to a firm. The firm receiving the loan has agreed to put up certain nonproducing assets for a value of \$200,000 as collateral. After 2 years, the amount still to be repaid is \$320,000. The bank estimates that for this type of loan, the probability of default is 6%, the recovery rate can be estimated at 50%, and the bank's analysts estimate that the collateral could be sold quickly for \$190,000. What is the LGD for the bank on this particular loan? What is the expected loss?

Answer. If we ignore the collateral, then we could use the first formula, i.e.,

$$LGD \text{ (dollar amount)} = 320,000 \times (1 - 50\%) = 160,000.$$

Considering the presence of collateral, instead we would find

$$LGD \text{ (as percentage)} = 1 - \frac{190,000}{320,000} = 40.63\%.$$

The expected loss has to employ the probability of default, that is:

$$\text{expected loss (dollar amount)} = PD \times LGD = 6\% \times 160,000 = 9,600.$$

Considering collateral, the expected loss can be computed as

$$\text{expected loss (dollar amount)} = 320,000 \times 6\% \times 40.63\% = 7,801.$$

It is often interesting and important to model the relationships between the variables just introduced. These may be highly nonlinear and may change over economic cycles. Hence, you should appreciate George Box's aphorism at this point.

Some investors may question the indication from rating agencies that are supposed to utilize very sophisticated models as the agencies could very well have a conflict of interest: if they give low ratings, the companies using their services may no longer do that, and therefore, exaggerating the ratings would be in the agencies' own interests.

As an investor in MBSs, one way to avoid the rating agencies' conflicts of interest would be to perform your own credit analysis (or have such analysis performed in house, i.e., within your own trustworthy firm). Another benefit of an independent analysis (if done well) could be the minimization of both model risk and the risk of financial loss due to bad decision making as a result of a bad model. Indeed, the main culprit in the subprime mortgage crisis, according to the readings, was the *failure of rating agencies' models to incorporate appropriate assumptions and valid inputs*. This is called model failure, and the risk of such failure is called "model risk." Since rating agencies' models were considered proprietary, investors could not verify their appropriateness and validity. A firm might be tempted to not get rated if they expect to receive a poor rating. However, smart investors see through this and quickly learn to assume that companies or securities with no rating have very low credit quality.

Certainly, constructing a good credit risk model for a huge pool of mortgages with widely varying characteristics (LTV and DTI ratios, borrower credit scores, loan purposes, etc.) is not simple. With that in mind, [read the article "Determination of Default Probability by Loss Given Default"](#) by Mišanková et al. to understand how the most important variables, such as loss given default (LGD) and probability of default (PD), are related. Also, consider carefully which of the inputs to these models can be known exactly and which ones need to be calculated, which may also require a model and thus implies model risk. Remember, no single model provides a complete or perfect interpretation of the world. Models cannot replace judgment that comes with true understanding. When a single investor relies blindly on a model, they will probably incur significant losses; when many investors make this mistake, the problem may become much larger, even systemic.

The math presented should not be difficult. Don't concern yourself too much with the technical discussion of the many variants of the fundamental formulae, though you should expect to learn about most of the regression approaches mentioned in later coursework. We will return to these formulae when we build out our own valuation model for a hypothetical MBS in the next course: Financial Data.

6. Regulation

Given the complexity of MBSs to model and value (which we will appreciate even more in the next course), investors have naturally relied on rating agencies' assessments. Actually, as you know from the readings, many participants in the mortgage and MBS market relied on other participants' truthfulness though they should not have. In markets where investors do not have enough information to make investment decisions in important markets (such as the housing market), the risks to that market may justify government intervention. This can take many forms, but regulation and bailouts are the most pertinent to a discussion of the financial crisis.

The first part of the "Erosion" deals with the many causes of the crisis. The section titled "Reluctant Regulators" lists multiple United States authorities that could have done more to prevent the crisis. As you read through the rest of this paper, consider how things may have turned out differently if regulations were in place.

For example, the section "Excessive Leverage" describes how the increased use of various derivatives added another layer of (synthetic) leverage. Could one of the regulators have put a limit on an investment bank's leverage? In fact, they did—after the crisis, as you read in the Dodd-Frank part of the reading. And what of the complex derivatives that enabled synthetic leverage? Yes, Dodd-Frank addressed this type of risk as well. As for the rating agencies' conflicts of interest (recall that they were paid by the issuers of the debt they were rating) and their hidden models, could regulators require that rating agencies be transparent about their modeling methodologies, as well as the "track record" (past performance statistics) of those models in predicting default? Yes—again, regulators addressed these issues in Dodd-Frank, as described in the reading.

But can all of the market's weaknesses and risks be simply regulated away? Read the short article "[Dodd-Frank Act: How Financial Reform May Be Going Wrong](#)" (Eisinger and Bernstein), which describes just a few of the very practical challenges that the post-crisis regulatory reform faced. Some of the practical difficulties described in this article might be overcome with time. Regardless, the debate about the proper scope of government intervention in the markets continues. For a sense of how these types of financial reforms are framed in very political (instead of pragmatic) terms, [read the report "Doomed to Repeat"](#) about the Dodd-Frank legislation as a response to the Great Financial Crisis. It discusses—also from its own very political perspective—the objectives of Dodd-Frank as well as the conservative criticisms of the legislation and provides a retort to these. Ask yourself as you read this if the issues could ever be resolved to both sides' satisfaction. Can the two sides of this argument ever agree?

Regulation, one of this week's themes, applies to every other principle we've encountered thus far. Credit risk, shorting, volatility, correlation, leverage, and nonlinearity all create opportunities for extreme outcomes and require regulation. As we are just scratching the surface of regulation, we will say much more, but for now, think of regulation as a first-order attempt to mitigate some of the extremeness.

Think back to our housing example: consider that income and home prices have some correlation, consider the nonlinearity of home equity, and consider the credit risk that banks have by lending a mortgage. With no money down, these principles combine to show that, if the economy worsens, people can lose their jobs, housing prices might drop, the home could be worth less than the original mortgage, the homeowner/borrower may no longer be able to pay, and the collateral (the home) itself might even be worth less than the loan. Regulation needs to be multifaceted to address the systemic risks in the financial ecosystem.

7. Conclusion

We applied basic and "high-level" credit analysis to a pool of mortgages that underly an MBS. We found that we didn't necessarily have all the credit information that we would like and that such information may not be reliable anyway due to market frictions and moral hazards. We noted that rating agencies face moral hazard in the form of conflicts of interest, and this is just one potential source of model risk. This prompted us to take our own first look at credit risk modeling and the calculation of probability of default. Since modeling risk is hard (and we will return to it in the Financial Data course), we asked ourselves whether moral hazard and other crisis-causing risk factors could just be regulated away. We learned that, in the United States, regulators attempted to accomplish this *after* the Great Financial Crisis. However, the effectiveness of these efforts are hotly debated along mostly ideological lines.

In the next lesson, we talk about the broader credit market to which MBS belongs. In the credit market, the main risk is credit risk, or the probability of default, so we look at that as well. The credit spread reflects credit risk, so we also talk about the relationship between the credit spread and the probability of default. As we discussed in this lesson, many investors rely on rating agencies for analysis of credit products like bonds. As such, in the next lesson, we also look at the relationship between credit ratings and probability of default levels. Another important risk, in credit markets as well as many other markets, is liquidity risk, so we look at ways to quantify this risk.

Footnotes

- The main credit scores in the U.S. are numbers in the range of 300-850, which are used to capture the credit quality of individual persons. A higher score suggests a more creditworthy borrower, and a lower credit score a less creditworthy borrower. Credit scores are very similar to credit ratings, which are assigned to companies or securities. One difference is that ratings range from AAA or Aaa (the best credit quality, implying a low probability of default) to C or D (very poor credit quality, implying a high probability of default).

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- Mišanková, Mária et al. "Determination of Default Probability by Loss Given Default." *Procedia Economics and Finance*, vol. 26, 2015, pp. 411–417. [10.1016/S2212-5671\(15\)00815-1](https://doi.org/10.1016/S2212-5671(15)00815-1).

Required Readings

Required Readings for this program are open access, which means you can access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. "Bonds." *Investor.gov*. www.investor.gov/introduction-investing/investing-basics/investment-products/bonds-or-fixed-income-products/bonds.
 - **Required pages:** All
 - **Estimated reading time:** 10 minutes
2. Booth, Anthony, and Boudewijn de Bruin. "Stakes Sensitivity and Credit Rating: A New Challenge for Regulators." *Journal of Business Ethics*, vol. 169, 2021, pp. 169–179. <https://link.springer.com/article/10.1007/s10551-019-04260-2>.
 - **Required pages:** pp. 169–173, 177–179
 - **Estimated reading time:** 20 minutes
3. "Corporate Bonds." *Investor.gov*. www.investor.gov/introduction-investing/investing-basics/investment-products/bonds-or-fixed-income-products.
 - **Required pages:** All
 - **Estimated reading time:** 10 minutes
4. "Introduction to Credit Risk Management." *TU Delft OpenCourseWare*. https://ocw.tudelft.nl/course-lectures/7-1-cds-cds-spreads/?course_id=13780.
 - **Required pages:** Entire article and video
 - **Estimated reading time:** 20 minutes
5. Sun, Xiaoli. *Liquidity, Credit Risk and Pricing of Corporate Bond*. 2007. Singapore Management University, Master's thesis. https://ink.library.smu.edu.sg/cgi/viewcontent.cgi?article=1025&context=etd_coll
 - **Required pages:** Chapters 1 and 2
 - **Estimated reading time:** 15 minutes
6. "What Are Corporate Bonds?" *Investor.gov*. https://www.sec.gov/files/ib_corporatebonds.pdf
 - **Required pages:** All
 - **Estimated reading time:** 10 minutes
7. "What Are High-Yield Corporate Bonds?" *Investor.gov*. <https://www.investor.gov/introduction-investing/general-resources/news-alerts/alerts-bulletins/investor-bulletins/what-are-0>.
 - **Required pages:** All
 - **Estimated reading time:** 10 minutes

NOTE: Content presented in the Required Readings and Lesson Notes will be tested in all assignments, including Quizzes.

LIQUIDITY AND THE CREDIT MARKET

Reading Time	75 minutes
Prior Knowledge	Bond, Issuance, Probability of Default (PD)
Keywords	Credit market, Credit rating, Credit Spread (CS), Credit Default Swap (CDS), Liquidity, Recovery Rate (RS), Probability of Default (PD), Bid-ask spread, Fungibility

In the last lesson, we applied the 5 Cs of credit analysis to see which of two MBS offerings had better credit qualities. We discussed how a mortgage exhibits nonlinearity, similar to other derivatives we have seen (call and put options). We witnessed how model risk contributed to the Great Financial Crisis. We discussed the impact of moral hazards (such as information asymmetry and conflict of interest) in the mortgage and bond markets and how this led to the need for new regulations such as Dodd-Frank in the U.S.

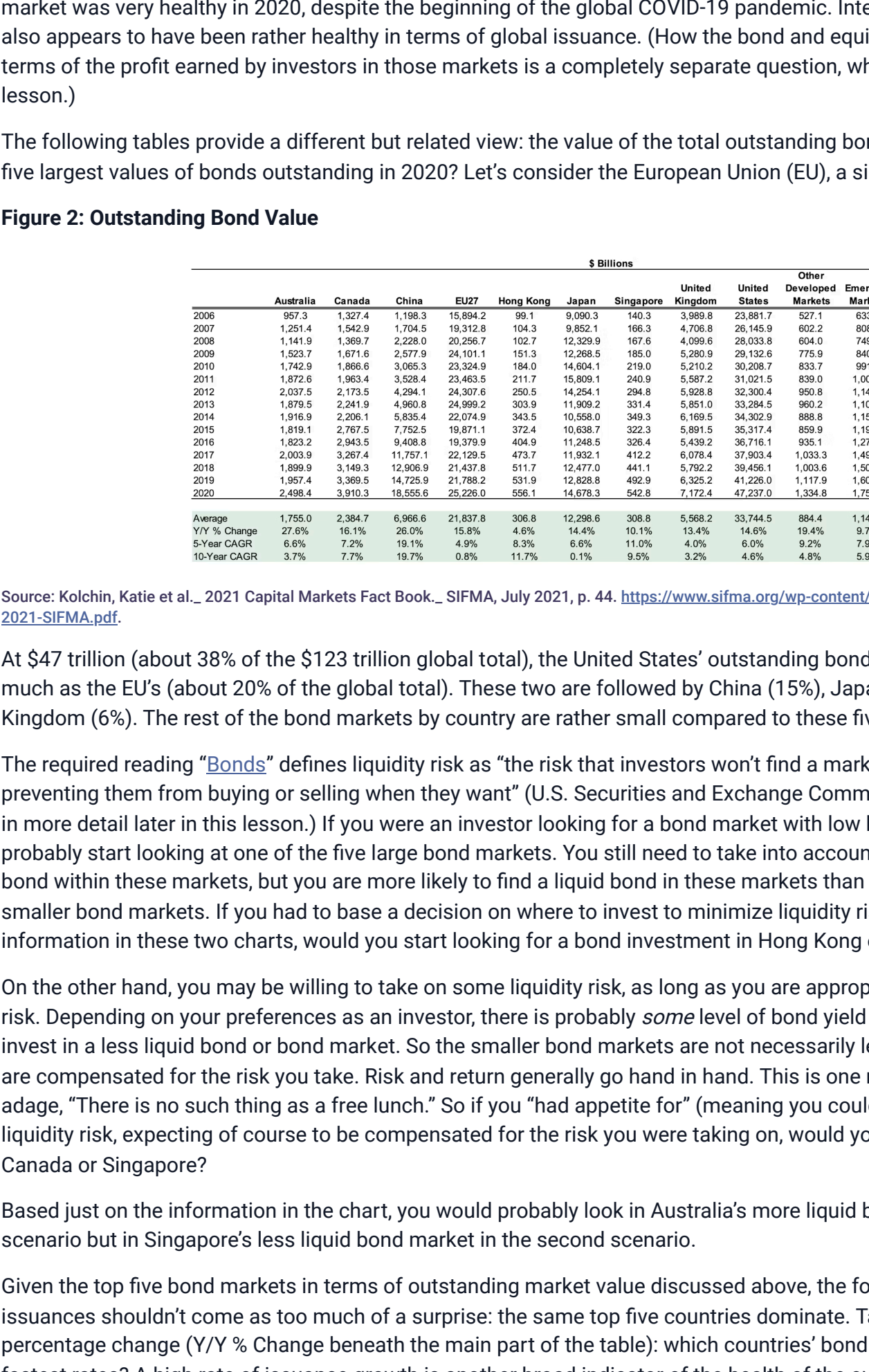
In this lesson, we quantify the size of the credit market, both overall as well as by country. We learn how to calculate a credit spread based on just a few inputs and how to “back out,” or calculate the implied probability of default from a bond’s credit spread. We discover the CDS (credit default swap) market as a way to trade the credit risk of a bond or bond issuer. We also learn the relationships between rating and probability of default and between rating and credit spread. We also discuss one of the big problems with certain risk measures and with credit ratings. Finally, we discuss some characteristics of liquidity and how we might estimate it.

1. The Credit Market and Its Components

The credit market includes securities that involve some sort of borrowing or debt. For this reason, the “debt market” is synonymous with the “credit market.” Since the debt market comprises mostly bonds, the credit or debt market may also be known as the bond market.

The bond market is vast, even compared to the global equity market. Compare the annual issuance of debt and equity in the following two graphs and be sure to note the units (\$T for bonds vs. \$B for equity). How many times larger was bond issuance in 2020 compared to equity issuance?

Figure 1: Bond and Equity Issuance



Source: Kolchín, Katie et al. 2021 *Capital Markets Fact Book*. SIFMA, July 2021, p. 38. <https://www.sifma.org/wp-content/uploads/2021/07/CM-Fact-Book-2021-SIFMA.pdf>.

Compare bond issuance in 2020 to 2018. What is this ratio? It appears to be exactly 1.5x. This suggests that the bond market was very healthy in 2020, despite the beginning of the global COVID-19 pandemic. Interestingly, the equity market also appears to have been rather healthy in terms of global issuance. (How the bond and equity markets *performed* in terms of the profit earned by investors in those markets is a completely separate question, which we do not address in this lesson.)

The following tables provide a different but related view: the value of the total outstanding bonds. Which countries had the five largest values of bonds outstanding in 2020? Let’s consider the European Union (EU), a single country for this purpose.

Figure 2: Outstanding Bond Value

	in \$trillions										
	Australia	Canada	China	EU27	Hong Kong	Japan	Singapore	United States	United Kingdom	Other Developed	Emerging
2020	185.5	269.9	1798.6	3,259.3	14.2	2,417.3	12.3	320.7	1,286.0	415.8	1,265.7
2019	185.1	261.1	1760.3	3,405.0	13.2	2,050.0	16.3	370.8	1,343.8	402.2	1,482.4
2018	179.7	362.0	1,624.7	3,432.3	10.7	2,430.4	21.0	1,115.1	4,051.1	384.0	872.7
2017	176.2	1,883.6	1,085.2	3,233.9	16.0	4,064.1	29.0	5,228.0	32,304.2	252.4	1,160.2
2016	300.0	472.8	76.8	3,607.3	30.2	3,584.8	37.8	474.4	6,777.2	470.4	1,753.5
2015	358.8	459.5	600.8	3,066.3	33.4	4,065.3	33.2	731.4	5,887.9	433.1	1,725.1
2014	346.7	516.9	391.5	3,727.2	40.9	4,203.3	43.8	734.7	6,007.0	541.7	1,911.9
2013	2,232.3	2,173.5	4,280.1	3,307.0	20.3	14,264.1	29.4	5,228.0	32,304.2	252.4	1,160.2
2012	311.6	597.2	4,058.3	3,406.9	38.3	3,197.7	43.8	440.2	5,055.2	580.3	1,783.9
2011	264.5	601.8	2,891.5	2,965.2	43.0	3,524.4	51.8	480.7	7,833.2	547.3	1,797.3
2010	268.3	600.7	2,790.0	2,968.5	44.2	3,250.5	38.0	474.1	6,841.4	547.3	2,284.7
2009	248.4	3,383.5	14,272.0	21,785.2	20.3	11,063.3	42.0	6,224.4	37,265.4	1,171.9	10,948.8
2008	216.1	508.8	2,780.1	2,965.1	48.8	3,125.2	30.2	416.3	6,816.0	274.1	2,427.4
2007	418.7	763.2	3,088.4	3,156.3	61.8	3,364.5	37.4	758.6	6,615.2	500.0	2,238.5
Average	277.4	476.0	1,364.3	3,410.4	43.5	3,603.0	36.2	724.8	6,231.0	332.2	1,811.7
Y/Y % Change	91.4%	37.8%	34.3%	28.9%	-27.7%	7.7%	-45.8%	80.8%	3.1%	25.3%	23.1%
5-Year CAGR	117%	8.1%	18.3%	2.8%	-6.9%	4.6%	12.0%	4.0%	8.0%	9.2%	7.9%
10-Year CAGR	2.7%	4.7%	21.8%	0.8%	-11.7%	0.1%	9.5%	3.2%	4.4%	4.8%	4.5%

Source: Kolchín, Katie et al. 2021 *Capital Markets Fact Book*. SIFMA, July 2021, p. 44. <https://www.sifma.org/wp-content/uploads/2021/07/CM-Fact-Book-2021-SIFMA.pdf>.

At \$47 trillion (about 38% of the \$123 trillion global total), the United States’ outstanding bond value is almost twice as much as the EU’s (about 20% of the global total). These two are followed by China (15%), Japan (12%), and the United Kingdom (6%). The rest of the bond markets by country are rather small compared to these five large ones.

The required reading “**Bonds**” defines liquidity risk as “the risk that investors won’t find a market for the bond, potentially preventing them from buying or selling when they want” (U.S. Securities and Exchange Commission). (We discuss liquidity in more detail later in this lesson.) If you were an investor looking for a bond market with low liquidity risk, you would probably start looking at one of the five large bond markets. You still need to take into account the liquidity of the specific bond within these markets, but you are more likely to find a liquid bond in these markets than in some of the other much smaller bond markets. If you had to base a decision on where to invest to minimize liquidity risk based only on the information in these two charts, would you start looking for a bond investment in Hong Kong or in Australia?

On the other hand, you may be willing to take on some liquidity risk, as long as you are appropriately compensated for that risk. Depending on your preferences as an investor, there is probably some level of bond yield that would induce you to invest in a less liquid bond or bond market. So the smaller bond markets are not necessarily less attractive, so long as you are compensated for the risk you take. Risk and return generally go hand in hand. This is one meaning of the economic adage, “There is no such thing as a free lunch.” So if you “had appetite for” (meaning you could tolerate and were seeking) liquidity risk, expecting of course to be compensated for the risk you were taking on, would you start your search in Canada or Singapore?

Based just on the information in the chart, you would probably look in Australia’s more liquid bond market in the first scenario but in Singapore’s less liquid bond market in the second scenario.

Given the top five bond markets in terms of outstanding market value discussed above, the following chart of annual issuances shouldn’t come as much of a surprise: the same top five countries dominate. Take note of the year-over-year percentage change (Y/Y % Change beneath the main part of the table): which countries’ bond markets are growing at the fastest rates? A high rate of issuance growth is another good indicator of the health of the overall credit market, though of course one still needs to evaluate bonds on an individual basis before investing. In this lesson, we will discuss, what bond markets seem to have been struggling over this 15-year period?

Figure 3: Global Bond Issuance

	in \$trillions										
	Australia	Canada	China	EU27	Hong Kong	Japan	Singapore	United States	United Kingdom	Other Developed	Emerging
2020	185.5	269.9	1798.6	3,259.3	14.2	2,417.3	12.3	320.7	1,286.0	415.8	1,265.7
2019	185.1	261.1	1760.3	3,405.0	13.2	2,050.0	16.3	370.8	1,343.8	402.2	1,482.4
2018	179.7	362.0	1,624.7	3,432.3	10.7	2,430.4	21.0	1,115.1	4,051.1	384.0	872.7
2017	176.2	1,883.6	1,085.2	3,233.9	16.0	4,064.1	29.0	5,228.0	32,304.2	252.4	1,160.2
2016	300.0	472.8	76.8	3,607.3	30.2	3,584.8	37.8	474.4	6,777.2	470.4	1,753.5
2015	358.8	459.5	600.8	3,066.3	33.4	4,065.3	33.2	731.4	5,887.9	433.1	1,725.1
2014	346.7	516.9	391.5	3,727.2	40.9	4,203.3	43.8	734.7	6,007.0	541.7	1,911.9
2013	2,232.3	2,173.5	4,280.1	3,307.0	20.3	14,264.1	29.4	5,228.0	32,304.2	252.4	1,160.2
2012	311.6	597.2	4,058.3	3,406.9	38.3	3,197.7	43.8	440.2	5,055.2	580.3	1,783.9
2011	264.5	601.8	2,891.5	2,965.2	43.0	3,524.4	51.8	480.7	7,833.2	547.3	1,797.3
2010	268.3	600.7	2,790.0	2,968.5	44.2	3,250.5	38.0	474.1	6,841.4	547.3	2,284.7
2009	248.4	3,383.5	14,272.0	21,785.2	20.3	11,063.3	42.0	6,224.4	37,265.4	1,171.9	10,948.8
2008	216.1	508.8	2,780.1	2,965.1	48.8	3,125.2	30.2	416.3	6,816.0	274.1	2,427.4
2007	418.7	763.2	3,088.4	3,156.3	61.8	3,364.5	37.4	758.6	6,615.2	500.0	2,238.5
Average	277.4	476.0	1,364.3	3,410.4	43.5	3,603.0	36.2	724.8	6,231.0	332.2	1,811.7
Y/Y % Change	91.4%	37.8%	34.3%	28.9%	-27.7%	7.7%	-45.8%	80.8%	3.1%	25.3%	23.1%
5-Year CAGR	117%	8.1%	18.3%	2.8%	-6.9%	4.6%	12.0%	4.0%	8.0%	9.2%	7.9%
10-Year CAGR	2.7%	4.7%	21.8%	0.8%	-11.7%	0.1%	9.5%	3.2%	4.4%	4.8%	4.5%

Source: Kolchín, Katie et al. 2021 *Capital Markets Fact Book*. SIFMA, July 2021, p. 37. <https://www.sifma.org/wp-content/uploads/2021/07/CM-Fact-Book-2021-SIFMA.pdf>.

At 92% and 81% year-over-year growth, the bond markets in Australia and the UK were healthy over the period in question according to this metric. By the same token, Singapore’s -26% “growth” suggests some problems in that market that require further investigation.

2. Credit Spreads and Credit Default Swaps

Again, the above assessments of the overall health of credit markets are much too broad to justify an actual specific investment. A more thorough analysis might start with the credit spread. A spread is just a difference in levels; that is, a **credit spread** is the difference in yield between a particular bond and the risk-free rate for the same maturity. Stated differently, the credit spread is the *yield premium* for a fixed maturity. A risky bond needs to yield more to attract investors. If a risk-free bond and a risky bond had the same yield, investors would only invest in risk-free bonds. Consequently, risky bonds need to yield more.

The riskier the bond, the higher the yield must be in order to attract investors. The difference in yield between a risky bond and a risk-free bond (the credit spread) is therefore a good indicator of the riskiness of the bond. In fact, the credit spread can be used to calculate a probability of default (PD) as discussed in the “**CDS and CDS Spreads**” reading and video.

As you watch the “Credit Default Swaps” video found on the same reading, pay special attention to the calculation of PD. (You may also find the transcript of the video below the embedding to be helpful.) Remember that the “recovery rate” is how much value the creditor expects to *recover* if the bond issuer (the borrower) defaults.

We have already considered the notion of recovery rate in the previous lesson, but it does not hurt to repeat it here as a further application. Recall the 4,000 Naira to Nushi, collateralized by her nice watch. Let’s say we estimate the watch to be worth 4,500 Naira. It would be irrational for Nushi to intentionally not repay her loan since the watch is worth more. But if she did default on (not repay) her loan, we have her watch, which we can sell to get some of our lost loan amount back. It may be the case that we aren’t able to find a buyer for the watch at the 4,500 Naira price. (Watches are not that *liquid*, as we will discuss more later in this lesson.) We may only be able to sell it quickly for 3,000 Naira, in which case we only recover 3,000 Naira out of our 4,000 Naira loan so that the recovery rate on this loan is 75%.

Once you are familiar with the PD calculation that uses credit spread (CS) and recovery rate (RR) as inputs from the video, you can calculate the probability of Nushi’s default with just a few assumptions. Let’s say you’d like to have a one-month loan term and the risk-free interest rate (RFR) is 1% per month. You decide to make Nushi the loan at the rate of 5% (so the bond yield is 5%); she should pay you back 4,200 Naira (4,000 Naira principal plus 200 Naira interest) in one month. Based on the definition of credit spread above, $CS = BY - RFR = 5\% - 1\% = 4\%$. We already calculated our recovery rate, $RR = 75\%$.

$$PD = CS / (1 - RR) = .04 / (1 - .75) = .16 = 16\%.$$

The probability of default implied by these assumptions is 16%. What if you actually believe Nushi’s PD is 25%? Then, the 4% CS is not enough to compensate you for the risk. You can easily rearrange the equation above to find the credit spread that corresponds to the PD you assign to Nushi:

$$CS = PD (1 - RR)$$

Now you would require a credit spread of

$$CS = PD (1 - RR) = .25 * (1 - .75) = .0625 = 6.25\%$$

So you would require a higher interest rate of $RFR + CS = 1\% + 6.25\% = 7.25\%$, or 290 Naira.

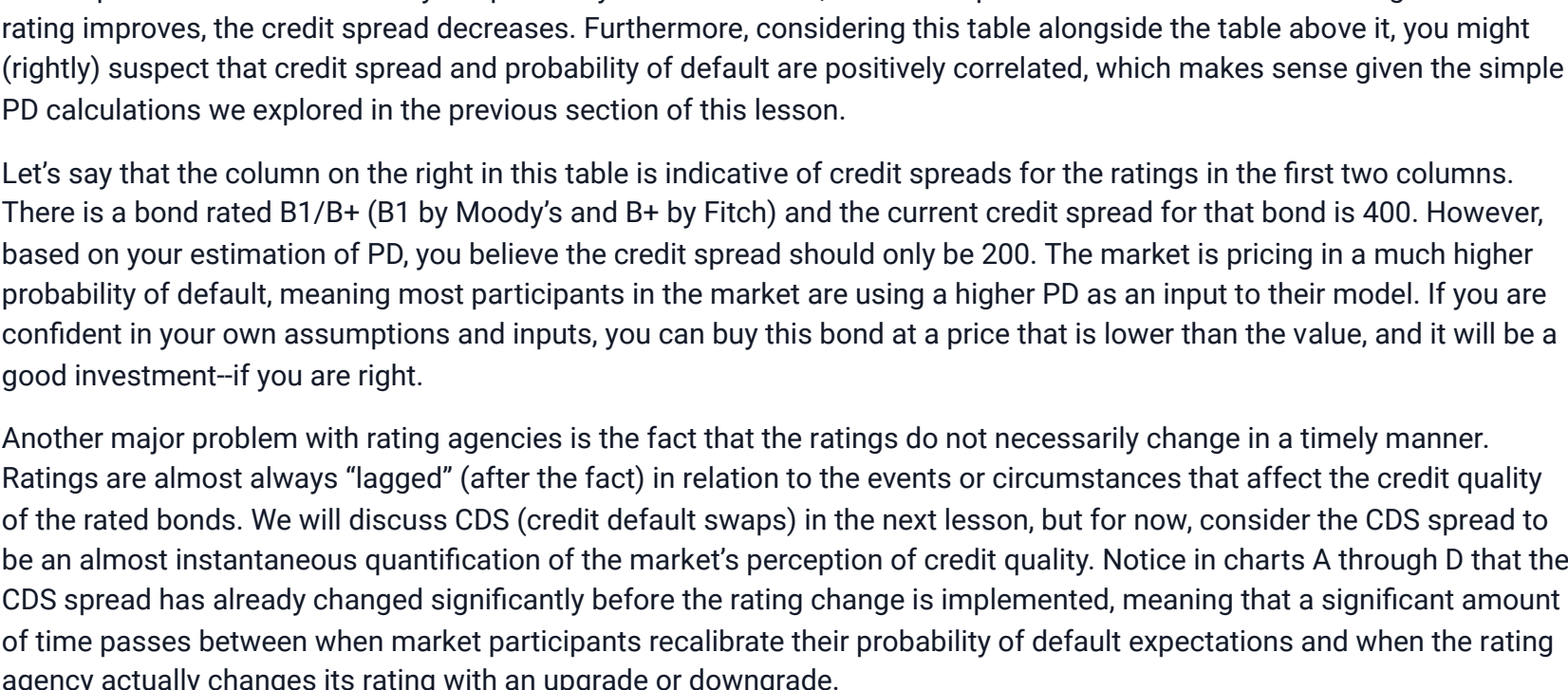
Would the CS go up or down if you believed that Nushi’s PD was lower, say only 10%?

$$CS = PD (1 - RR) = .10 * (1 - .75) = .025 = 2.5\%, \text{ which is lower than the original CS} = 6.25\%.$$

Go ahead and manipulate the inputs in the following estimation formula and see whether the impact of the changes are linear or nonlinear.

What did you find? Discuss with your classmates in the forum.

Probability of Default, Credit Risk, Credit Spread



IMPORTANT REMARK about the PD formula. Say that $RFR = 2\%$, the interest rate on the bond is 5%, so that $CS = 3\%$. Assume now that $RR = 90\%$. If we use the formula $PD = CS / (1 - RR)$, we find

$$PD = \frac{3\%}{1 - 90\%} = 0.30$$

which seems to be a very reasonable conclusion. However, if in the same setting, we change RR from 0.90 to 0.99, we find

$$PD = \frac{3\%}{1 - 99\%} = 3.1.$$

This clearly does not work since a probability cannot be larger than 1. So what is going on? It is very simple: the formula for PD only works locally. When we assume that CS was 3% with a RR of 90%, that could have been correct, but it is very unlikely that CS = 3% when RR = 99%. In fact, one could even argue whether we need to have a credit spread/premium when, in case of default, we are able to recover nearly the entire investment.

The gist of things here is that we cannot toy with the formula completely at will because the CS and RR inputs are not independent, but correlated. Therefore, do not include $PD \approx 3!$ which would show a complete ignorance of the laws of probability, not a good trait to exhibit for someone who wants to be a quantitative analyst.

As discussed in “**7.1 CDS and CDS Spreads**,” a CDS is very similar to traditional insurance in that the CDS buyer pays a premium to insure against the risk of a bond’s default, much like an insurance customer buys (pays a regular premium for) insurance against the risk of, for example, a catastrophe damaging a house.

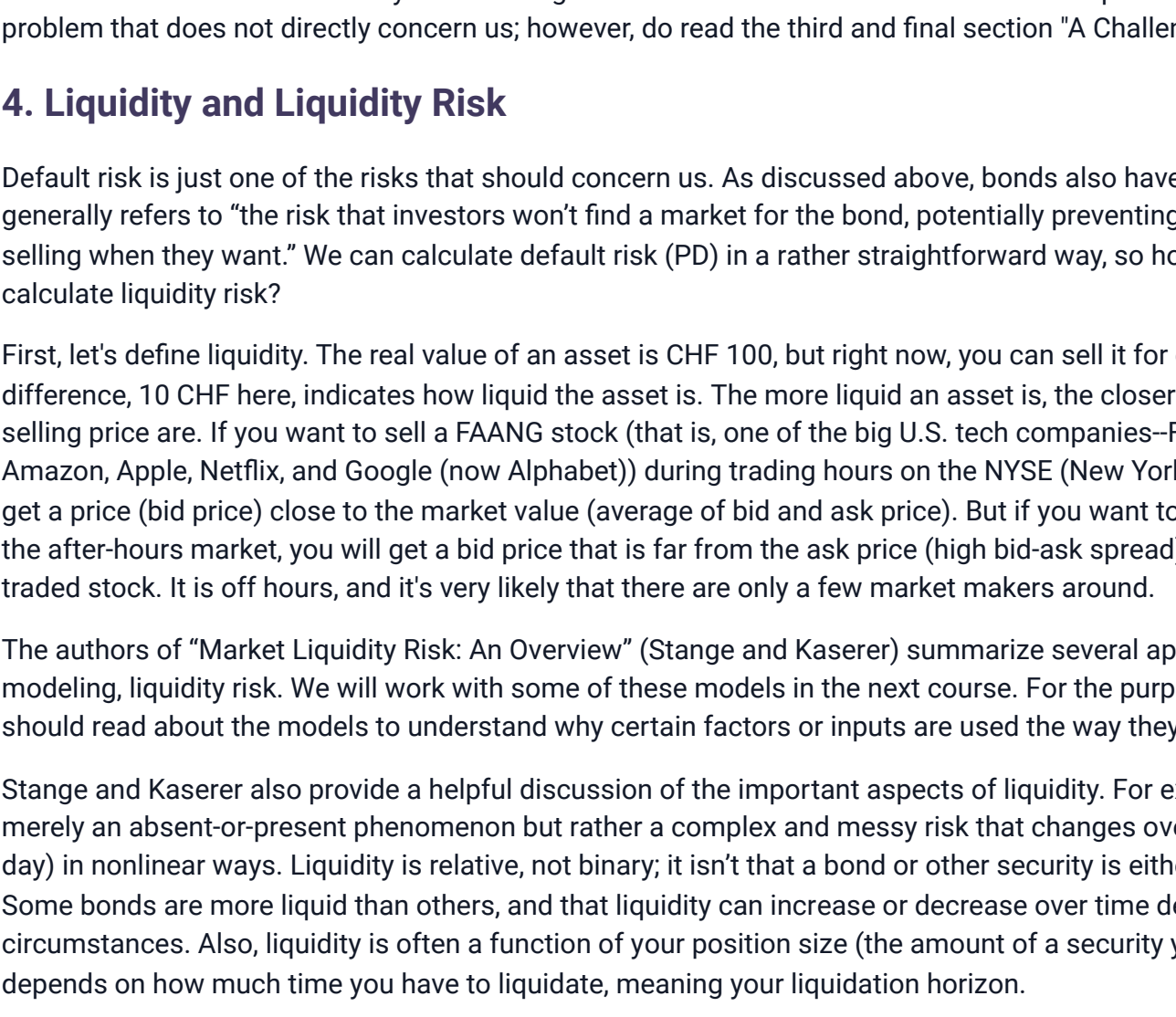
An important difference between a CDS and a traditional insurance, however, is that you don’t need to own any bonds to buy a CDS. You can buy protection on something you don’t even own. In other words, you can use a CDS to *speculate* on a bond instead of just *hedging* the risk on the bond. Likewise, you can use a CDS to sell protection on something.

This is very different from the traditional insurance market. If someone else could buy fire insurance on your house, they could make a lot of money if your house burned down. They could make so much money that they might even set your house on fire. This would be an example of moral hazard (where one party takes excessive risk for their own benefit because they will not bear the full cost of those risks”), which we discussed in the last lesson.

When you own a bond and buy CDS protection on that bond, you have the same interest rate risk from the bond that you had before, but you’ve offloaded the credit risk to some other market participant. If you don’t have the bond but you believe that the bond is headed for default, you can buy protection (on the bond that you *don’t* have) and make money on the default.

This aspect of CDSs is analogous to buying a put option on an equity security (stock): you don’t need to own the stock in order to buy a put or sell a call on that stock.

Figure 4: Credit Default Swap: Protection Buyer, Protection Seller, and Reference Entity



CDSs therefore allow for the easy buying and selling of the credit risk of a specific borrower. We say that CDSs “commodify” credit risk: CDSs allow us to buy and sell credit risk as if it were a mere commodity. In other words, CDSs nicely package, price, and transfer credit risk from one party to another, turning it into a commodity.

By using this **Shiny App**, students will have the opportunity to plot the CDS curves of several large market cap assets on several dates.

3. Ratings and Credit Spread

As we saw in the calculation examples above, higher PD implies a higher credit spread. Ratings are a function of the PD, or probability of bankruptcy as used in the table: the highest rating (AAA) is for bonds that have almost no risk of default, meaning PD at or very near zero.

Note that “investment grade” refers to bonds rated BBB or better, and bonds rated below BBB are called “junk.” There is nothing disparaging about the latter term; it is just credit market jargon. Another term for these sub-BBB bonds is “high yield.”

Figure 5: Ratings and Probabilities of Default

	Credit rating	Probability of bankruptcy, %
AAA	AAA	0.01
A+	A+	0.07
A	A	0.66
A-	A-	2.5
BBB	BBB	7.54
BB	BB	16.63
B+	B+	25.0
B	B	36.8
B-	B-	45.0
CCC	CCC	59.01
C	C	70.0
D	D	80.0
		100.0

Required Readings

Required Readings for this program are open access, which means you can access them at no cost. The link provided in the citation will take you directly to the reading or to a page where you can download it.

1. Adelino, Manuel et al. "The Role of Housing and Mortgage Markets in the Financial Crisis." *Annual Review of Financial Economics*, vol. 10, 2018, pp. 25–41. https://dspace.mit.edu/bitstream/handle/1721.1/122348/ARFE_Housing_v15.pdf?sequence=2&isAllowed=y.

- **Required pages:** All
- **Estimated reading time:** 40 minutes

2. Dong, Kai et al. "The Dynamic Correlation among Financial Leverage, House Price, and Consumer Expenditure in China." *Sustainability*, vol. 13, no. 5, 2021, pp. 1–18. <https://www.mdpi.com/2071-1050/13/5/2617>.

- **Required pages:** pp. 1–7, 14–16
- **Estimated reading time:** 15 minutes

3. Kling, Arnold. "'Skin in the Game': A Political Treatise for Liberty that Turns the Golden Rule Upside-Down." *Foundation for Economic Education*, 20 May 2018. <https://fee.org/articles/skin-in-the-game-a-political-treatise-for-liberty-that-turns-the-golden-rule-upside-down/>.

- **Required pages:** All
- **Estimated reading time:** 10 minutes

4. "Leverage: What It Is and How We Can Profit from It." *SimpleStockInvesting.com*. <http://www.simplestockinvesting.com/leverage.htm>.

- **Required pages:** All
- **Estimated reading time:** 10 minutes

5. Schoen, Edward J. "The 2007–2009 Financial Crisis: An Erosion of Ethics: A Case Study." *Journal of Business Ethics*, vol. 146, 2017, pp. 805–830. <https://doi.org/10.1007/s10551-016-3052-7>.

- **Required pages:** All
- **Estimated reading time:** 35 minutes

6. Yuksel, Asli. "The Relationship between Stock and Real Estate Prices in Turkey: Evidence around the Global Financial Crisis." *Central Bank Review*, vol. 16, no. 1, 2016, pp. 33–40. <https://www.sciencedirect.com/science/article/pii/S130307011600007X>.

- **Required pages:** All
- **Estimated reading time:** 10 minutes

7. All, Ghulam et al. "Do House Prices Influence Stock Prices? Empirical Investigation from the Panel of Selected European Union Countries." *Economic Research-Ekonomska Istraživanja*, vol. 30, no. 1, pp. 1840–1849. <https://www.tandfonline.com/doi/full/10.1080/1331677X.2017.1392882>.

- **Required pages:** All
- **Estimated reading time:** 25 minutes

NOTE: Content presented in the Required Readings and Lesson Notes will be tested in all assignments, including Quizzes.

LEVERAGE AND CRISIS

Reading Time	30 minutes
Prior Knowledge	Debt-to-Income (DTI) ratio, Loan-to-Value (LTV) ratio, Collateral, Interest rate, Moral hazard
Keywords	Correlation, Wealth effect, Credit-price effect, Leverage

In the last lesson, we compared equity and the credit market in terms of overall size. We also quantified the relationship between bond yield and probability of default. We learned about CDS and credit ratings as indicators of credit risk (in terms of probability of default). We also discussed ways of assessing and quantifying liquidity.

In this lesson, we observe an example of correlation in the real world, namely the stock and real estate markets (especially in Turkey). We observe the role of leverage in this correlated relationship as well as the risks of leverage in individual transactions (e.g., trading stock "on margin," buying a house with a mortgage, etc.) as well as in the larger market (widespread increases in DTI). The latter led to the Great Financial Crisis, which required governmental response in the form of regulation.

1. Correlation, House Prices, and the Stock Market

We have discussed the strong relationships between many factors or variables, for example, between probability of default and credit spread: when one increases, we expect the other to increase. We can measure the strength of this relationship in many ways, but one of the most basic is correlation: the strength (and direction) of a "linear" relationship. There is a linear relationship between two variables when the graph of the two variables more or less fits on a straight line. Note: We use Pearson correlation in this lesson, which is considered parametric, meaning it assumes the data have particular distributions. Spearman and Kendall correlations do not make assumptions about the distribution of the data, but these alternative correlations still measure the association between two variables, i.e., the degree to which two variables move in the same direction or opposite directions.

With your fresh and thorough understanding of the concept of correlation, read the introduction and literature review of the required reading paper, "The Relationship between Stock and Real Estate Prices in Turkey: Evidence around the Global Financial Crisis." Yuksel discusses two ways that we might understand the relationship between the stock and real estate markets: the wealth effect and the credit-price effect. In both views, there is a positive correlation between prices in the two markets. There also appears to be a positive linear relationship between these markets and the economy (both globally and in Turkey) as measured by GDP, as you might expect.

According to economists, changes in a consumer's wealth cause changes in the amounts and distribution of one's consumption. People typically spend more overall when one of two things is true:

- People actually get richer when they get a job that pays a higher salary; or
- People perceive themselves to be richer: for example, the assessed value of their home increases, or a stock they own goes up in price.

Note the distinction between actual and perceived additional wealth. There is often a psychological component about feeling rich. We may have bought a house for, say, \$500,000 and now we see that we can sell it for \$600,000, so we feel richer. However, one could counter that if they sell the house, their only house, they would likely need to buy a new one which would also very likely have gone up in price, so the additional \$100,000 is not exactly free purchasing power. However, many consumers cannot see this, and they feel rich. On the other hand, someone may own several homes: one to live in and others as investments. In this case, the investor is experiencing an actual wealth effect.

Housing wealth effects are widely believed to have played an important role in the boom of the early 2000s and the recession that followed. As a matter of fact, the 2000s saw a large run-up and subsequent decline in aggregate house prices that led housing to play an unusually large role in driving the business cycle over that period.

According to the "wealth effect" mechanism, in a rising stock market, would you prefer to buy or to sell a house? It's a good time to sell your house because the increased demand for real estate means you can get a better price. It's a bad time to buy because prices are increasing.

According to the credit-price effect, rising real estate prices should help the stock market by allowing companies to borrow more (to invest in their own operations) and at cheaper rates. Say you wanted to invest during a time of rising interest rates in only one of two very similar companies in the same industry: "Big Prop Inc." (BPI), which has significant real estate holdings, or "NoMaD Inc." (NMDI) with no real estate assets. The credit-price effect suggests that BPI can borrow more or at cheaper rates than NMDI because BPI has more collateral to secure its loans. (Recall that collateral is one of the 5 Cs of credit analysis.) You would certainly want to take into account that BPI may have lower financing costs than NMDI when you analyze the two companies in more detail. On the other hand, say real estate prices have just started to decrease after years of rising. Now, you might be concerned about whether BPI is prepared for the resulting decrease in the property values that collateralize their loans, while NMDI's financing costs may not change as drastically because it didn't have as much debt in the first place.

1.1 Leverage Risk

As we see in the example of Big Prop Inc., borrowing has advantages and disadvantages. In that example, BPI was able to borrow more to invest in its operations and perhaps make more profit, but it was also vulnerable to additional risks on top of the risks of its main operations: in this case, real estate price risk. As you read the required reading article "Leverage: What It Is and How We Can Profit from It," keep track of the various potential dangers and disadvantages of leverage that are mentioned.

Note:

1. The article mentions closed-end funds (CEFs) and exchange-traded funds (ETFs). These are distinct investment types, but in both types of these funds, you can buy and sell shares on an exchange, similar to traditional stocks. An investor in either of these fund types owns a portion of the underlying portfolio of securities. And, importantly, either may entail **leverage**, as described in the article.
2. This article also mentions swaps, which will be covered in much more detail in later courses. For now, just start to appreciate that certain financial products can enable very large amounts of **leverage**.

Now that you are familiar with correlation and leverage, you are ready to read "The Role of Housing and Mortgage Markets in the Financial Crisis." Notice you are already familiar with many of the mortgage-related credit terms, such as DTI (debt-to-income) and LTV (loan-to-value) ratios, from previous lessons. Indeed, you will read how "banks lent against increasing collateral values" (Adelino et al. 1) (which we recognize as the "wealth effect") and "underestimated the risk of defaults" (Adelino et al. 1), which we recognize as model risk. At the same time, there may be other technical terms that will make more sense as you make progress through the courses in this program.

As you read this paper:

1. Reinforce your understanding of the inherent relationship between borrowing and leverage. Note too that there are two types of leverage at play in this article: the leverage implied by buying a home with a low (or no) down payment and the leverage implied by DTI (debt to income, where debt is just another term for borrowing).
2. Use the 5 Cs of credit analysis to categorize the many factors that led to massive mortgage defaults.
3. Find as many examples of moral hazard as you can, whether or not they were explicitly discussed in the previous lesson.

2. Regulation and Crisis

Given the many types of moral hazard that proliferated at the time, it appears that the mortgage and housing markets were in need of some additional laws and stricter punishments for not adhering to the laws that were already in place. The above article (Adelino et al.) suggests that appropriate regulation could control or at least mitigate many of the factors that contributed to the Great Financial Crisis (GFC), and this echoes many of the arguments we read in a previous lesson from Schoen's "The 2007–2009 Financial Crisis: An Erosion of Ethics: A Case Study." The latter article also details many aspects of the damage from the GFC, most of which did not impact the major decision makers in the markets themselves, meaning that those players were able to take excessive risks without bearing the costs of those risks. Kling summarizes Nassim Nicholas Taleb's position in the introduction to his review of Taleb's *Skin in the Game: Hidden Asymmetries in Daily Life* as follows: "Only when people are, themselves, exposed to the adverse consequences of their choices do they take risks that are constructive for society. When they do not have 'skin in the game,' they take risks that are harmful and dangerous."

Extensive regulation was needed due to the fact that the market was not able to address this large-scale moral hazard. Recall from Module 4, Lesson 4 that taking out a mortgage has a similar risk profile to having a call option, both of which are non-linear. The potential gains are significant (theoretically, infinite), but the risk of loss is very limited. Under such circumstances, you might be tempted to buy as many call options (or houses, using mortgages) as you possibly can because someone else has most of the risk.

Since the market did not develop ways to manage these moral hazards on its own, the government needed to step in with regulations. The market needed new "rules of the game": regulations that would stop incentivizing (i.e., rewarding with handsome profits) risky behavior. As you read the article, make note of the activities and behaviors of market participants that could be considered unfair, which in turn demanded regulators step in. Also, keep track of the consequences of these actions and behaviors at the level of individuals as well as the financial system itself and society as a whole.

To balance this pro-regulation viewpoint, read this required reading, a review of Nassim Nicholas Taleb's book *Skin in the Game*. Taleb became the world's most famous risk manager with his 2007 book, *The Black Swan*. Like Schoen, Taleb argues that moral hazard threatens society's best interests. The big difference is that Taleb sees regulation as a centralized restriction of freedom, which actually releases people from responsibility. In Kling's paraphrase of Taleb, "freedom and progress both depend on decentralization, which realigns decision-making with responsibility." While Schoen sees centralized rule-making (regulation) as the antidote to the moral hazards that evolve in financial markets, Taleb sees regulation as providing an excuse for market participants to behave unethically, so long as their actions are technically legal.

What do you think? Now that you are familiar with the incentives and behaviors that led to the Great Financial Crisis and have read about the arguments for and against financial regulation, join the forum and suggest some guidelines for regulation in the mortgage market that might help avoid similar catastrophes in the future, while ensuring the proper functioning of the mortgage market. Alternatively, propose to your peers what the proper function of financial markets is in the first place. Please also respond to your fellow students' suggestions with your own thoughtful and constructive feedback.

3. Conclusion

First, we tried to unravel the correlation between the stock and real estate markets, and we saw that leverage helped explain this correlation. We discussed how leverage can be very helpful or very dangerous. New regulations were required to combat the dangerous combination of leverage and moral hazard. In the next module, we take a deeper look at leverage by first defining it in terms related to the balance sheet. We also talk about various other forms of real estate finance: in relation to the real estate developer, the real estate investment trust (REIT), the landlord, and the renter. We also take a look at some innovative alternatives and supplements to the traditional home mortgage.

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